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Design and synthesis of *s*-triazine-Isatin hybrids with potent anticancer activity, targeting A549 lung adenocarcinoma *via* EGF inhibition

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ABSTRACT

Merging *s*-triazine and isatin *via* a hydrazone linkage an efficient strategy to access new potential anticancer derivatives. Consequently, herein we report a new library of fifty new compounds based on isatin-*s*-triazine hydrazone derivatives. We used a straightforward orthogonal synthetic pathway to prepare disubstituted *s*-triazine hydrazine derivatives, whose basic structure was modified and then condensed with substituted isatin at the C-5 position to afford the new library of isatin-*s*-triazine hydrazone derivatives. The newly synthesized compounds were fully characterized by ^{1}H NMR, ^{13}C NMR and elemental analysis. The synthesized compounds were evaluated for their antiproliferative activity against lung cancer cell line (A549), and safety profile against Human Caucasian foetal lung (WI-38). The results revealed that all tested compounds exhibited cytotoxic effects against A549 when compared to the reference drug sorafenib. Most of the tested compounds exhibited safety indices that were either equivalent to or higher than sorafenib's index of 11.025, highlighting their superior or comparable safety profiles. Notably, compounds **9a**, **9d**, **10d**, **13a**, **13d**, **14a-14e**, **15a**, **15b**, **15d**, **15e**, and **11a-11e** exhibited particularly promising safety profiles. However, in terms of selectivity, compounds **6a-6e**, **7a-7e, 8a-8e**, **10d**, **12d**, **13a**, **13d**, **14a**, **14d** (with the highest selectivity index), **11c**, and **11d**, **14e**, **15a**, **15b**, **15d**, and **15e** showed greater potency than sorafenib, as evidenced by selectivity indices exceeding 1. In addition, all compounds demonstrated broad-spectrum anti-trypsin properties, five compounds (**7c**, **8a**, **12d**, **13a**, and **13d**) exhibited promising anti-trypsin effects at their anticancer IC_{50} values (97.912 ± 12.14, 93.511 ± 10.32, 99.623 ± 14.71, 98.411 ± 25.3, and 75.123 ± 4.32, respectively), while **7c**, **8a**, **12d** and **13a** showed the highest inhibitory effect, surpassing the inhibitory effect of bis(5-amidino-2-benzimidazolyl)methane (IC_{50} = 56.617 ± 2.14). When compared to the inhibitory effect of rivaroxaban (53.223 ± 0.98) at a concentration of 0.0013 μM against Factor Xa, all compounds showed lower inhibitory effects except for **6b**, **6c**, **7a**,**7e**, **8a**, **8b**, **8c**, **8e**, **9b**, **9c**,

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9d, **10a**, **11a**, **12a**, **13a**, and **14a** showed the highest inhibitory effect. Furthermore, all compounds acted as epidermal growth factor (EGF) inhibitors to varying degrees. Compared to the IC_{50} of Sorafenib (68.2 % inhibitory effect), several compounds exhibited potentially greater potency as EGF inhibitors, with more potent inhibitory effects at their IC_{50} values compared to sorafenib. These compounds include **7c**, **10b**, **10c**, **10d**, **10e**, **12d**, **12e**, **14b**, **14c**, **15a-15e**, **11b**, **11d**, and **11e**. All the compounds that demonstrated superior EGF inhibition *in vitro* were selected and upon docking showed both higher docking scores and free binding energies (−9.728: 6.890 & −51.49: 40.74, respectively) compared to Sorafenib −4.649 and −34.96 respectively.

1. Introduction

Cancer is considered as the main public health problem in the modern world, and remains one of the leading causes of death among developed countries [1]. Lung cancer still the leading cause of cancer death, with an estimated 1.8 million deaths (18 %), followed by colorectal (9.4 %), liver (8.3 %), stomach (7.7 %), and female breast (6.9 %) cancers [2].

Numerous chemotherapeutic drugs are commercially available, and several others are in clinical trials. However, several serious side effects produced during the treatment with these drugs. Another problem is that despite cytotoxic effect *in vitro* and inhibit tumor growth *in vivo* additional complications may arise because of the existence of small subtype of cells called cancer stem cells (CSCs). Such cells are relatively resistant to therapy and can after treatment with the cytotoxic drug has ended, effect repopulation with cancer cell *in vivo*. [3] Regardless of the unlimited progress undertake in cancer treatment, several limitations are still present. [4,5] Hence, the search for new compounds that are both effective and selective is still needed and will be a challenge in therapeutic chemistry.

The epidermal growth factor receptor "EGFR" is a member of tyrosine kinases (TKs) which are considered the major causes of several types of cancer. [6,7] EGFR is also one of the major tumor markers in numerous cancer types, where its signaling in tumors, as opposed to

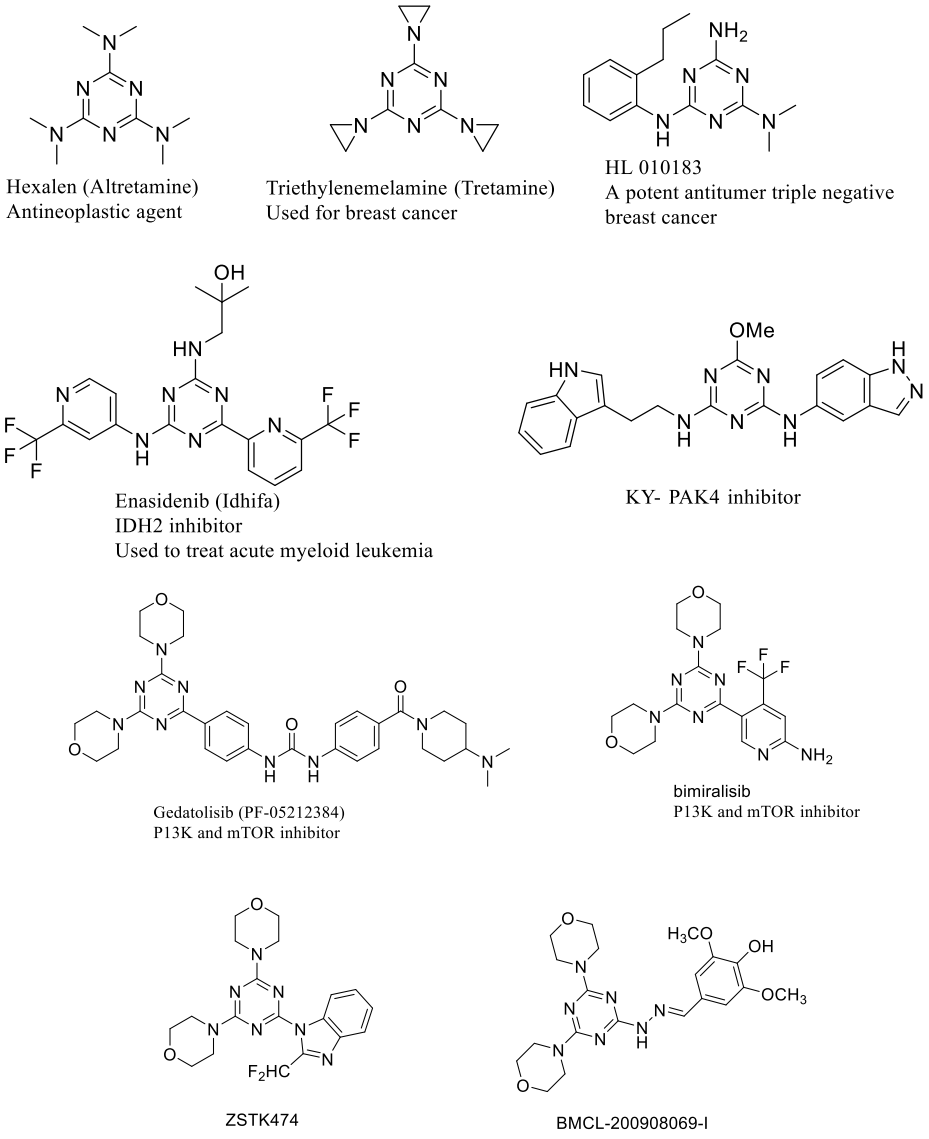


Fig. 1. Examples of some anticancer drugs bearing *s*-triazine moiety.

normal cells, becomes nonregulated, resulting in EGFR overexpression and/or obtaining a gain-of-function mutation [8–12]. This performance is considered the major cause of tumor cell proliferation, attacking the surrounding tissues and resulting in an increase in angiogenesis [13]. Accordingly, disconnecting EGFR communicating signals is considered to be one of the major targets to attack tumors caused by its mis--regulation.[8–14]

The 1,3,5-triazine "*s*-triazine" possesses an auspicious status in the field of drug discovery, with numerous bioactive molecules [15–18]. It is resent in many naturally occurring compounds as well as commercially available drugs like tretamine (first approved by U.S. FDA in 1990), gedatolisib [19,20], HL 010183 [20], enasidenib [21], altretamine [22], KY [19] and bimiralisib [23] (Fig. 1).

Also, there are several anticancer 1,3,5-triazines that have reached phase III clinical trials such as ZSTK474 [24] and BMCL- 200908069-1 [25] that proved efficacy against different cancer cell lines (Fig. 1). The wide interest in the 1,3,5-triazine core has led to synthesis of a large number of derivatives with in some cases promising anticancer activities [25–28].

Likewise, it has been reported that the combination of *s*-triazine with other important bioactive moieties results in a notable improvement of their potential activity in many biological and pharmaceutical applications [29–32].

On the other hand, isatin, is well known compound for more than a century and still being used extensively in medicinal chemistry [33–36]. In recent years, number of isatin derivatives were reported with extensive biological activities [37–41], as well as topo II inhibitor [42]. Recently, novel hydrazone-isatin derivatives have been reported as potential epidermal growth factor receptor (EGFR) [43,44] (Fig. 2), also showed that this compound possesses drug-likeness properties, showed a significant decrease of the cell population in the G2/M phase and led to a significant increase in early and late apoptosis, comparable to erlotinib.

Compounds comprising hydrazone derivatives with a conjugated structure (−NH–N═CH−) are appealing target compounds for novel therapeutic development because of their potentially varied biological effects [45–50]. It has been proposed that the inhibitory effect of these hydrazones is due to kinase inhibition [51–53] or the formation of radicals and decrease of the mitochondrial membrane potential [54].

In accordance with the results mentioned above and as an aspect of our continuing efforts [17,55–59] to create effective anticancer agents-based combination of *s*-traizine moiety and isatin ring *via* hydrazone linkage.

In the present work the design of the target compounds relied on the use of the hybrid pharmacophore approach to design and synthesize new series of *s*-triazine-based hybrids through merging the pharmacophoric parts of isatin with different substituent at 5-position and *s*-triazine with different substituent at 4,6-position in one compound framework with the primary objective of developing potent anticancer agents against A549 cell line as well as WI-38. Accordingly, a library of 50 new *s*-triazine-isatine hydrazone-based derivatives were designed (Fig. 3) based on the reported data, where bismorpholine *s*-triazine derivatives were reported as scaffold contributes significantly to the resistance of cancer cells [60–63]. In addition, substituted aniline *s*-traizine derivatives showed promising results against A549 cells and Epidermal Growth Factor Receptor tyrosine kinase [26–28,64]. According to the previously reported data morpholine, piperidine and substituted aniline attached to *s*-traizine ring [25–28] have potential anticancer activity. In addition, substituted isatin with different groups such as Cl, Br or F as electron withdrawing group and CH_3 as electron donating group could enhance their pharmacological effects [43]. In the present work, we rational designed and synthesized small library contained 50 compounds as shown in Fig. 3, and their activity was tested against WI-38 and A549. Furthermore, a molecular docking study was performed.

2. Results and discussion

2.1. Chemistry

The new series of *s*-triazine-isatin hydrazone derivatives were synthesized following the reported method, where hydrazinyl-*s*-triazine derivatives were reacted with isatin derivatives in the presence of drops of acetic acid and ethanol as solvent [56]. The initial step involved the nucleophilic substitution of the chlorine atom of cyanuric chloride **1** by different amines (morpholine, piperidine, aniline, or *p*-substituted aniline) at 0–5 °C to afford the products 2,4-dichloro-6-substituted-*s*-triazine derivatives, **2a-j**. The second step involved the replacement of the second chlorine atom by a second amine at room temperature for 12–24 h, yielding 2-chloro-4,6-disubstitued-*s*-triazine derivatives **3a-j**. Compounds **3a-j** were reacted with hydrazine hydrate (80 %) under reflux in ethanol for 8–12 h [58–61] to afford **4a-j**. Compounds **4a-j** were then reacted with isatin derivatives **5a-e** in the presence of catalytic amount of acetic acid and ethanol as solvent to afford the target products (**6a-e) - (15a-e) (**Scheme 1). The structure of the new synthesized compounds was confirmed by NMR (^{1}H- and ^{13}C-), and elemental analysis (Supporting *information, **SI1-100***).

2.2. Biology

2.2.1. Triazine compounds induced cytotoxicity in human cancer cells

The new synthesized *s*-triazine-isatin hydrazone derivatives were classified into 10 groups (**GPI-GPX,** Fig. 4) according to the type of substituent attached to the triazine moiety and each group includes five different compounds according to the substituent at 5-position in isatin ring as shown in Fig. 4. Consequently, the fifty new *s*-triazine-isatin hydrazone derivatives, where they are classified according to the substituent at *s*-triazine and isatin moieties. Additionally, the safety index (SI), selectivity, percentage inhibition of trypsin, percentage inhibition

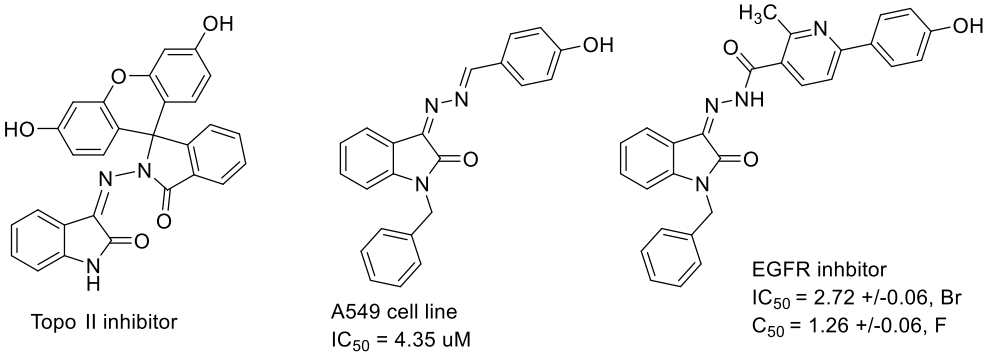


Fig. 2. Topo II and Isatin-hydrazone derivatives as potential epidermal growth factor receptor.

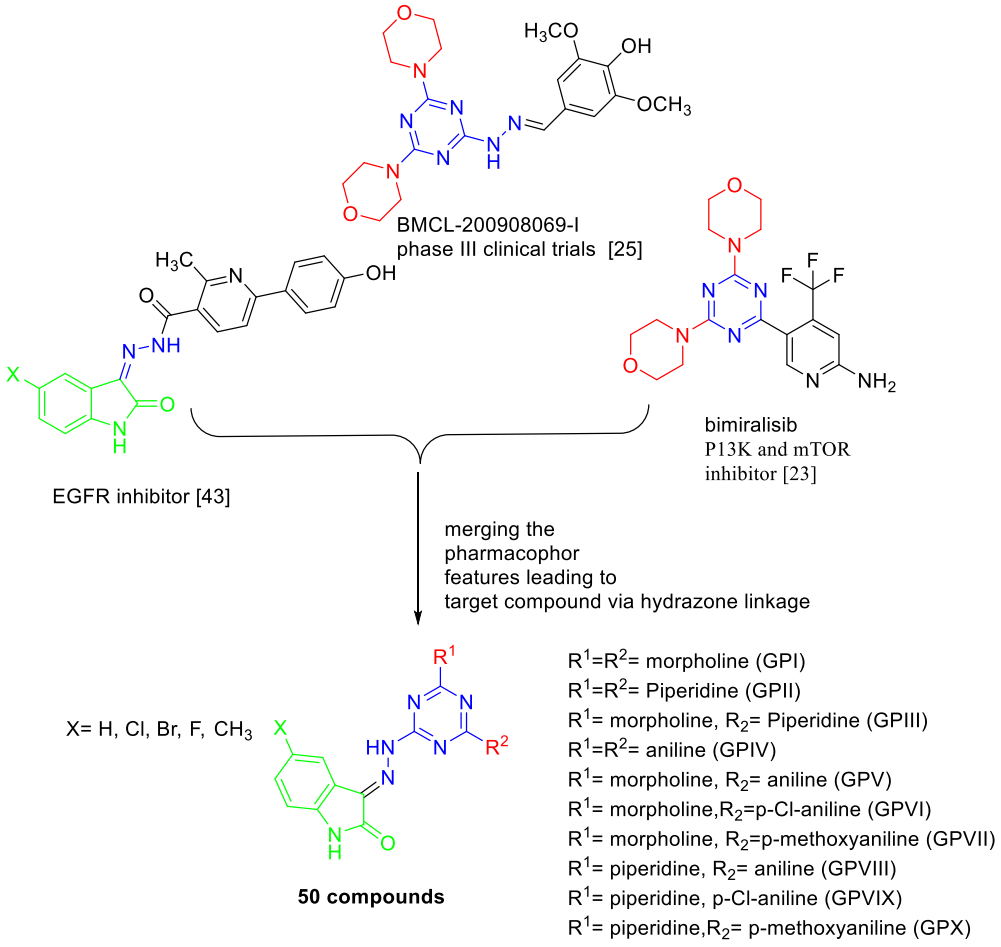
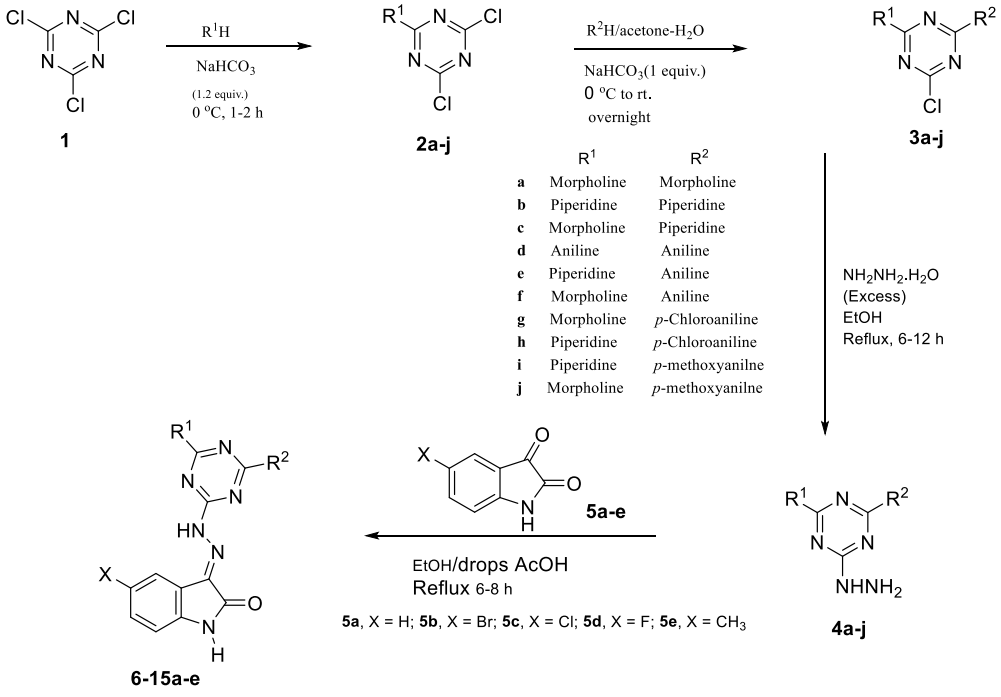
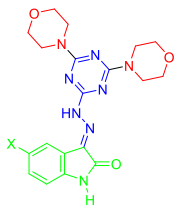


Fig. 3. Selected isatin and *s*-triazine derivatives as anticancer agent and our rational designed 50 compounds.



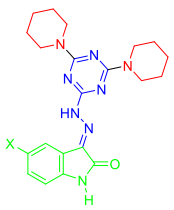
Scheme 1. Synthetic route for synthesis of *s*-triazine-satin hydrazone derivatives.

GPI



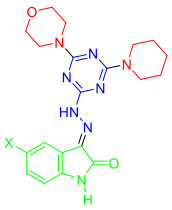
6a, X = H; **6b,** X = Br; **6c,** X = Cl; **6d,** X = F; **6e,** X = CH_3

GPII



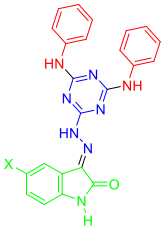
7a, X = H; **7b,** X = Br; **7c,** X = Cl; **7d,** X = F; **7e,** X = CH_3

GPIII



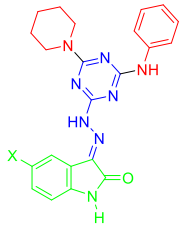
8**a,** X = H; **8b,** X = Br; **8c,** X = Cl; **8d,** X = F; **8e,** X = CH_3

GPIV



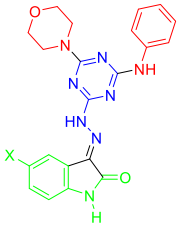
9a, X = H; **9b,** X = Br; **9c,** X = Cl; **9d,** X = F; **9e,** X = CH_3

GPV



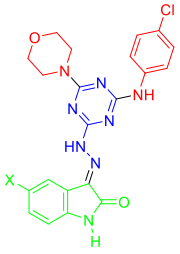
10a, X = H ; **10b,** X = Br; **10c,** X = Cl; **10d,** X = F; **10e,** X = CH_3

GPVI



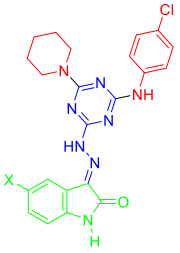
11a, X = H ; **11b,** X = Br; **11c,** X = Cl; **11d,** X = F; **11e,** X = CH_3

GPVII



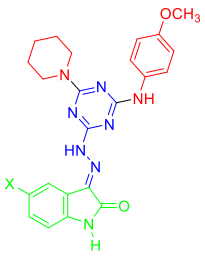
12a, X = H ; **12b,** X = Br; **12c,** X = Cl; **12d,** X = F; **12e,** X = CH_3

GPVIII



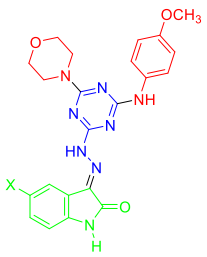
13a, X = H ; **13b,** X = Br; **13c,** X = Cl; **13d,** X = F; **13e,** X = CH_3

GPIX



14a, X = H ; **14b,** X = Br; **14c,** X = Cl; **14d,** X = F; **14e,** X = CH_3

GPX



15a, X = H ; **15b,** X = Br; **15c,** X = Cl; **15d,** X = F; **15e,** X = CH_3

Fig. 4. Structures of the target compounds-based *s*-triazine-isatin hydrazone derivatives.

of Factor Xa, and percentage inhibition of epidermal growth factor (EGF) were evaluated at their IC_{50} values, as detailed in Table 1.

The results revealed that all tested compounds exhibited cytotoxic effects against WI-38 and A549. Their IC_{50} values within the range between 0.105 ± 0.01 and 0.383 ± 0.03 μM against A549 were lower than those against WI-38, where the IC_{50} values within the range between 0.098 ± 0.01 and 3.583 ± 0.06 μM, indicating their relative safety, when compared to the reference drug sorafenib (IC_{50} values for WI-38 and A549 are 2.15 ± 0.01 and 0.195 ± 0.02 μM, respectively). Most of the tested compounds demonstrated either equivalent or superior activity and safety indices, with indices equal to/or greater than sorafenib's index of 11.025. Notably, compounds **9a** (11.621 ± 1.039 μM), **9d** (11.671 ± 1.11 μM), **10d** (12.087 ± 0.53 μM), **11a-e** (11.733 ± 0.93, 11.129 ± 0.23, 11.731 ± 3.75, 13.211 ± 1.12, 11.245 ± 0.25 μM, respectively), **13a** (11.469 ± 1.47 μM), **13d** (12.785 ± 1.05 μM), **14a-e** (12.312 ± 2.13, 11.404 ± 0.64, 11.268 ± 1.27, 14.651 ± 2.81, 13.55 ± 0.72 μM, respectively), **15a** (13.319 ± 0.99 μM), **15c** (12.481 ± 0.72 μM), **15d** (12.073 ± 1.59 μM), and **15e (**12.844 ± 0.71 μM) exhibited particularly promising safety profiles. However, in terms of selectivity, compounds with the highest selectivity index **6a-e**, **7a-e**, **8a-e**, **9d**, **10d**, **13a**, **13d**, **14a**, **14d**, **11b**, **11d, 14e**, **15a**, **15b**, **15d**, and **15e**, showed greater potency than sorafenib (Table 1), as evidenced by selectivity indices exceeding 1. In addition, all compounds demonstrated broad-spectrum anti-trypsin properties, five compounds (**7c**, **8a**, **12d**, **13a**, and **13d**) exhibited promising anti-trypsin effects at their anticancer IC_{50} values (97.912 ± 12.14, 93.511 ± 10.32, 99.623 ± 14.71, 98.411 ± 25.3, and 75.123 ± 4.32 μM, respectively), while **7c**, **8a**, **12d** and **13a** showed the highest inhibitory effect, surpassing the inhibitory effect of bis(5-amidino-2-benzimidazolyl)methane (IC_{50} = 56.617 ± 2.14 μM). When compared to the inhibitory effect of rivaroxaban (53.223 ± 0.98 μM) at a concentration of 0.0013 μM against Factor Xa, a majority of the compounds showed lower inhibitory effects except for compounds **6b** (55.689 ± 0.92 μM), **6c** (58.568 ± 1.82 μM), **7a** (68.631 ± 3.72 μM), **7e** (60.657 ± 1.827 μM), **8a** (74.693 ± 3.72 μM), **8c** (74.342 ± 1.82 μM), **8b** (55.532 ± 2.42 μM), **8e** (56.547 ± 1.32 μM), **9b** (65.897 ± 2.92 μM), **9c** (85.689 ± 4.52 μM), **9d** (70.347 ± 3.82 μM), **10a** (60.647 ± 1.52 μM), **11a** (57.006 ± 1.61 μM), **12a** (68.231 ± 7.41 μM), **13a** (55.036 ± 5.11 μM), and **14a** (72.032 ± 1.31 μM), which showed highest inhibitory effect compared to the standard rivaroxaban as shown in Table 1.

Regarding SAR, the compounds in **GPI-GPX** as shown Fig. 4 and Table 1, where **GPI** showed better activity against both cell lines WI-38 and A549 compared to the standard Sorafenib. Of note that all substituted isatin in this group (**GPI**, **6b-e**) showed less EGR than unsubstituted **6a** and the standard drug Sorafenib. While compounds **6b** (5-bromoisatin) and **6c** (5-chloroisatin) showed the % Factor Xa higher than the standard Rivaroxaban, while **6d** (5-fluoroisatin) and **6e** (5-methylisatin) showed lower % Factor Xa in this group.

GPII showed almost the same as **GPI** and better activity against both cell lines WI-38 and A549 compared to the standard drug Sorafenib. Compound **7a** unsubstituted isatin and **7e** (5-methylisatin) gave the higher % Factor Xa compared to the standard Rivaroxaban and their analogs *bis*-morpholine **6a** and **6e**. Compound **7b** (5-bromoisatin) gave higher activity against EGF (75.467 ± 3.62 μM) compared to *bis*-morpholine **6b (**46.471 ± 1.42 μM) and the standard Sorafenib (68.251 ± 5). Also compound **7c** showed superior % Trypsin inhibition (97.912 ± 12.14 μM) compared to *bis*-morpholine **6c** (5.479 ± 0.52 μM) and the standard *bis*(5-amidino-2-benzimidazolyl)methane (56.617 ± 2.14 μM).

GPIII, where piperidine ring was replaced by one morpholine gave better results than compared to *bis*-piperidine group (**GPII**). Where Compound **8a** (unsubstituted satin) showed the best activity against trypsin inhibition while **8a**, and **8c** showed better activity in % Factor Xa compared to the analogs in **GPII 7a** and **7c** (**8a, 8c** *vs* **7a,7c)**, also showed better activity compared to the standard Rivaroxaban. Compound **8d** (5-fluroisatin) showed the best value for selectivity 1.98 with the lowest EGF (50.471 ± 1.82 μM) compared to the other derivatives in the same group and its analogs in **GPI** and **GPII**.

GPIV, where the two morpholine rings (**GPI**) and the two piperidine rings (**GPII**) replaced by two aniline rings showed lower IC_{50} for both cell lines (WI-38 and A549) but similar to the standard drug Sorafenib, with selectivity less than 1 and lower EGF compared to the **GPI, GPII**, and **GPIII**. While compound **9a** unsubstituted isatin and **9d** (5-methylisation) derivative showed higher SI (11.621 ± 1.039 μM, and 11.671 ± 1.11) compared to the analogous in the **GPI, GPII**, and **GPIII** and the standard Sorafenib (11.025 ± 0.93 μM**).** Also, Furthermore, compound **9e** (5-methylisation) gave almost similar safety index (11.073 ± 1.11) to the standard. Also, **9b** and **9c** (5-bromoisatin and 5-chloroisatin) showed lower SI than the standard but still better than the analogs in **GPI, GPII**, and **GPIII**.

In term of % of Factor Xa, **9b**, **9c**, and **9d** showed higher value at their IC_{50} compared to standard Rivaroxaban and their analogs in **GPI**, GPII and GPIII.

GPV, the presence of the piperidine ring with the aniline ring increase the activity against WI-38 and A549 and EGR inhibition compared to the analogs *bis*-aniline derivatives in **GPIV**, and the standard Sorafenib. Compound **10e** (5-methylisatin) showed the best EGF inhibition (83.613 ± 6.12 μM) compared to its analogs in another group (**GPI-GPVI**) and the standard drug Sorafenib. While compound **10d** showed the best SI (12.087 ± 0.53) compared to the standard and other its analogs in another group (**GPI-GPVI**) with EGF inhibition (79.132 ± 3.52 μM) which is also better than the standard Sorafenib, while it did not show any activity against trypsin and Factor Xa.

GPVI, where one morpholine ring with one aniline showed almost the same as its analogs in **GPV** in term of higher activity against EGF with higher value for SI, compound **11e** showed the highest activity (85.128 ± 6.82) at its IC_{50} compared to others and the standard drug Sorafenib.

GPVII, were the presence of the chlorine in aniline moiety decreased the activity compared to it analogs in **GPVI** (unsubstituted aniline), while compound **12a** unsubstituted isatin gave better value for Factor Xa compared to the standard and compound **12d** (5-fluroisatin) derivative showed superb values for Trypsin inhibition (99.623 ± 14.71 μM) which is the best in all tested fifty compounds. Also, **12b (**5-bromoisatin) and **12d (**5-fluoroisatin) derivatives showed better activity against EGF compared their analogs in **GPVI** and the standard Sorafenib (**12b**, **12d** *vs* **11b, 11d,** and Sorafenib).

GPVIII showed less activity compared to its analogs **GPVII** while compound **13a** unsubstituted isatin derivative gave better values for SI, trypsin and Factor Xa compared to the standard Sorafenib.

Replacement of the chlorine atom in the aniline ring (**GP VIII**) by OCH_3 in **GPIX** improved the SI and selectivity value but it did not improve the activity against the two cell lines WI-38 and A549 but still better than the standard drug. Also, it improves the activity against EGF especially compound **14b** (5-bromoisatin) derivative and **14c** (5-chloroisatin) derivative *vs* standards Sorafenib and its analogs **13b** and **13c**.

Replacement of the chlorine atom in the aniline ring for (**GP VII**) by OCH_3 for **GPX** improved the SI, selectivity and the activity against the two cell lines WI-38 and A549. Also, it improves the activity against EGF for all series and showed better activity compared to its analogs **GPVII** and the standards Sorafenib but it showed lower activity against trypsin and Factor Xa compared to others and the standard drug used in this study.

In conclusion, the SAR studies indicated that the type of substituents on s-triazine and isatin moieties has a significant effect on the activity against WI-38 and A549. In addition, they play the major effect on selectivity index (SI), selectivity, the % of Trypsin inhibition, the % of Factor Xa, and the % of Epidermal growth factor inhibition.

2.2.2. Molecular docking and interactions with receptor amino acid residues

All the compound that demonstrated superior EGFR inhibition *in vitro* were selected and upon docking showed both higher docking scores and free binding energies (−9.728: −6.890 & −51.49: −40.74,

Table 1
Antiproliferative activity of s-triazine-isatin hydrazine derivatives **6a-e- 15a-e** against WI-38 and A549; Selectivity index (SI), Trypisn inhibition, Factor Xa, and Epidermal growth factor inhibition.

GP	Cpd	WI-38 IC_{50} (μM)	A549 IC_{50} (μM)	Safety index (SI)	Selectivity	% Trypsin inhibition at compound IC_{50}	% Factor Xa inhibition at compound IC_{50}	%Epidermal growth factor inhibition at compound IC_{50}
GPI	**6a**	0.971 ± 0.04	0.105 ± 0.01	9.293 ± 0.37	1.857	4.794 ± 0.04	50.032 ± 2.31	65.056 ± 1.82
	6b	1.149 ± 0.06	0.162 ± 0.01	7.078 ± 0.69	1.203	19.863 ± 1.11	55.689 ± 0.92	46.471 ± 1.42
	6c	1.064 ± 0.06	0.135 ± 0.01	7.902 ± 0.65	1.444	5.479 ± 0.52	58.568 ± 1.82	62.075 ± 2.81
	6d	1.602 ± 0.07	0.169 ± 0.01	9.452 ± 0.57	1.154	21.233 ± 0.91	25.124 ± 0.82	59.634 ± 3.52
	6e	1.098 ± 0.05	0.137 ± 0.01	8.021 ± 1.09	1.42	9.931 ± 0.73	47.451 ± 3.42	42.547 ± 2.62
GPII	**7a**	1.336 ± 0.09	0.178 ± 0.02	7.498 ± 0.66	1.096	13.013 ± 0.33	68.631 ± 3.72	59.137 ± 4.22
	7b	1.265 ± 0.09	0.185 ± 0.01	6.852 ± 0.73	1.054	8.562 ± 0.08	14.053 ± 0.72	75.467 ± 3.62
	7c	1.216 ± 0.08	0.164 ± 0.01	7.4301 ± 0.89	1.189	97.912 ± 12.14	26.631 ± 1.22	56.643 ± 4.14
	7d	0.938 ± 0.06	0.098 ± 0.01	9.488 ± 1.26	1.989	25.342 ± 0.19	52.634 ± 2.72	53.237 ± 3.82
	7e	1.354 ± 0.05	0.191 ± 0.02	7.091 ± 0.35	1.021	17.808 ± 0.17	60.657 ± 1.82	62.374 ± 5.42
GP III	**8a**	1.105 ± 0.08	0.147 ± 0.01	7.529 ± 1.51	1.326	93.511 ± 10.32	74.693 ± 3.72	62.872 ± 5.92
	8b	1.203 ± 0.04	0.192 ± 0.01	6.256 ± 0.36	1.016	10.958 ± 1.1	55.532 ± 2.42	65.723 ± 3.51
	8c	1.173 ± 0.09	0.175 ± 0.01	6.701 ± 1.27	1.114	8.2191 ± 0.08	74.342 ± 1.82	63.648 ± 4.62
	8d	0.889 ± 0.04	0.098 ± 0.01	8.992 ± 0.96	1.989	ND	ND	50.471 ± 1.82
	8e	1.058 ± 0.04	0.141 ± 0.01	7.491 ± 0.54	1.383	6.506 ± 0.54	56.547 ± 1.32	68.637 ± 7.12
GP IV	**9a**	2.391 ± 0.150	0.206 ± 0.01	11.621 ± 1.03	0.947	ND	5.224 ± 0.12	40.321 ± 2.42
	9b	2.917 ± 0.08	0.273 ± 0.02	10.703 ± 0.48	0.714	6.849 ± 0.4	65.897 ± 2.92	40.368 ± 1.42
	9c	2.919 ± 0.14	0.267 ± 0.02	10.917 ± 0.64	0.731	0.342 ± 0.01	85.689 ± 4.52	45.341 ± 3.83
	9d	2.334 ± 0.12	0.199 ± 0.01	11.671 ± 1.11	0.981	41.781 ± 1.3	70.347 ± 3.82	38.317 ± 0.93
	9e	2.598 ± 0.09	0.235 ± 0.01	11.073 ± 1.11	0.829	ND	0.565 ± 0.05	32.396 ± 0.82
GPV	**10a**	2.640 ± 0.13	0.247 ± 0.02	10.671 ± 0.81	0.789	39.383 ± 1.82	60.647 ± 1.52	55.016 ± 2.61
	10b	2.545 ± 0.08	0.249 ± 0.01	10.229 ± 0.56	0.783	ND	ND	75.637 ± 4.93
	10c	2.469 ± 0.09	0.235 ± 0.02	10.493 ± 0.45	0.829	18.151 ± 0.97	49.309 ± 2.41	71.708 ± 3.83
	10d	1.680 ± 0.06	0.139 ± 0.01	12.087 ± 0.53	1.403	ND	ND	79.132 ± 3.52
	10e	2.981 ± 0.12	0.296 ± 0.03	10.054 ± 0.49	0.659	ND	ND	83.613 ± 6.12
GP VI	**11a**	2.518 ± 0.11	0.215 ± 0.01	11.733 ± 0.93	0.907	1.435 ± 0.07	57.006 ± 1.61	40.631 ± 4.11
	11b	2.676 ± 0.05	0.241 ± 0.02	11.129 ± 0.23	0.809	3.874 ± 0.14	30.078 ± 3.12	64.892 ± 2.61
	11c	2.282 ± 0.16	0.195 ± 0.01	11.731 ± 3.75	1	5.363 ± 0.31	26.035 ± 2.52	70.078 ± 3.82
	11d	1.742 ± 0.11	0.132 ± 0.01	13.211 ± 1.12	1.477	45.890 ± 3.52	28.089 ± 0.81	71.078 ± 3.23
	11e	2.772 ± 0.05	0.247 ± 0.02	11.245 ± 0.25	0.789	1.923 ± 0.23	36.356 ± 1.52	85.128 ± 6.82
GP VII	**12a**	2.234 ± 0.09	0.216 ± 0.01	10.331 ± 0.87	0.903	35.274 ± 1.22	68.231 ± 7.41	55.607 ± 2.32
	12b	3.583 ± 0.06	0.383 ± 0.03	9.345 ± 0.19	0.248	24.657 ± 1.6	27.369 ± 1.21	70.142 ± 4.82
	12c	2.623 ± 0.18	0.267 ± 0.02	9.823 ± 0.79	0.731	12.328 ± 0.09	50.953 ± 3.52	67.314 ± 6.12
	12d	1.865 ± 0.11	0.1701 ± 0.01	10.942 ± 1.08	1.164	99.623 ± 14.71	38.231 ± 1.52	75.934 ± 7.11
	12e	2.290 ± 0.11	0.229 ± 0.01	9.998 ± 0.84	0.852	11.301 ± 0.17	39.321 ± 1.32	40.463 ± 3.82
GP VIII	**13a**	2.061 ± 0.12	0.179 ± 0.01	11.469 ± 1.47	1.089	98.411 ± 25.3	55.036 ± 5.11	56.314 ± 3.41

(*continued on next page*)

Table 1 (*continued*)

GP	Cpd	WI-38 IC_{50} (μM)	A549 IC_{50} (μM)	Safety index (SI)	Selectivity	% Trypsin inhibition at compound IC_{50}	% Factor Xa inhibition at compound IC_{50}	%Epidermal growth factor inhibition at compound IC_{50}
	13b	2.803 ± 0.13	0.268 ± 0.01	10.446 ± 1.12	0.728	ND	1.923 ± 0.01	40.053 ± 2.96
	13c	2.881 ± 0.14	0.279 ± 0.01	10.301 ± 1.52	0.699	ND	0.984 ± 0.08	45.014 ± 2.42
	13d	1.458 ± 0.09	0.114 ± 0.01	12.785 ± 1.05	1.711	75.123 ± 4.32	26.023 ± 1.22	65.314 ± 5.42
	13e	2.877 ± 0.14	0.276 ± 0.01	10.427 ± 1.41	0.707	ND	ND	53.315 ± 1.92
GP IX	**14a**	2.138 ± 0.16	0.174 ± 0.01	12.312 ± 2.13	1.121	0.685 ± 0.04	72.032 ± 1.31	40.678 ± 2.83
	14b	2.618 ± 0.13	0.229 ± 0.02	11.404 ± 0.64	0.852	20.890 ± 1.42	34.063 ± 1.52	70.047 ± 6.12
	14c	2.558 ± 0.09	0.227 ± 0.01	11.268 ± 1.27	0.859	ND	38.012 ± 3.21	83.098 ± 8.12
	14d	1.297 ± 0.06	0.089 ± 0.02	14.651 ± 2.81	2.191	39.041 ± 1.54	39.079 ± 2.47	62.396 ± 5.92
	14e	1.491 ± 0.03	0.109 ± 0.04	13.55 ± 0.72	1.789	11.644 ± 0.92	35.178 ± 1.92	46.053 ± 3.42
GPX	**15a**	1.364 ± 0.05	0.102 ± 0.01	13.319 ± 0.99	1.912	3.943 ± 0.04	53.078 ± 2.34	73.987 ± 6.82
	15b	2.461 ± 0.08	0.231 ± 0.02	10.682 ± 0.43	0.844	0.227 ± 0.01	3.608 ± 0.02	70.008 ± 5.12
	15c	1.504 ± 0.08	0.121 ± 0.011	12.481 ± 0.72	1.612	0.141 ± 0.01	1.821 ± 0.43	83.784 ± 4.62
	15d	1.811 ± 0.09	0.150 ± 0.01	12.073 ± 1.59	1.321	4.109 ± 0.06	37.089 ± 1.41	80.089 ± 3.91
	15e	1.430 ± 0.06	0.111 ± 0.01	12.844 ± 0.71	1.756	1.224 ± 0.21	5.752 ± 2.14	72.089 ± 2.32
Sorafenib		2.15 ± 0.01	0.195 ± 0.02	11.025 ± 0.93	–			68.251 ± 5.93
Rivaroxaban (0.0013 μM)							53.223 ± 0.98	
Standard[a]						56.617 ± 2.14		

[a] Bis(5-amidino-2-benzimidazolyl)methane (4.4 μM).

respectively) compared to Sorafenib −4.649 and −34.96 respectively. After Docking the 2D and 3D interactions were obtained (Fig. 5), all selected compounds showed H-bonding with both MET769 and ASP831 with exception of **11e, 11d, 7c** which formed H-bonds with MET769 & LYS721 and MET769 only respectively. Additionally, **14b, 14c, 10e, 15e, 15b, 15d,** and **12d** showed extra H-bonding with LYS721. All these compounds maintained hydrophobic interaction with LEU694 and LEU820 while some of them could achieve other hydrophobic interactions in the active region of EGFR; with **11e** showing the broadest coverage of hydrophobic interactions with binding site residues (Table S1, *Supplementary Data*).

Additionally, potential binding of compounds that showed superior *in vitro* activity inhibiting Factor Xa and Thrombin than rivaroxaban and bis(5-amidino-2-benzimidazolyl) methane respectively were investigated. Regarding Factor Xa all the compounds showed higher docking scores and free binding energies ranging (−6.955: −4299 & −64.77: −33.89), respectively) along with L-shaped binding involving interactions with S1 and S4 pocket residues similar to that of rivaroxaban with exception of **6b, 6c,** and **10b**. Compounds maintained hydrophobic van der Val interactions with one or more of hydrophobic S4 pocket residues, along with hydrogen bonding with residue 216 and/or 214 located in S1 pocket with exception of **11a**. Most of the compounds maintained hydrophobic interactions with one or more residues of S1 pocket and π-π stacking interactions with one or more of S4 pocket residues with exceptions shown in Table S2 (*Supplementary data*) Compounds **13a, 9b,** and **9c** showed hydrogen bonding with 195 a member of catalytic triad and oxyanion hole and 61 located near the catalytic triad. Finally, compound **9d** showed a π-πstacking interaction with 222 located in allosteric site.

For investigating potential binding of selected compounds with thrombin, two crystalline structures were used, porcine trypsin PDB:1S85 and human thrombin PDB:1KTS (Tables S2 and S3, Supplementary Data). For trypsin the five compounds showed multiple hydrophobic interactions with three or more of the main substrate binding pockets "S1 pocket" residues, with docking scores and free binding energies (−5.082: −3.861 & −52.54: −32.63, respectively). Compounds **13d, 13a,** and **12d** formed π-π stack bond with His57 with and without additional π--Cation bond, respectively. While **8a** and **7b** formed H-bond with Ser195 and Gly193, respectively. Finally, **13d, 13a,** and **12a** showed halogen bond formation with Asp189. On the other hand, the same five compounds were docked against thrombin structure PDB:1KTS. Compound **13a,12d** and **12a** showed H-bond formation with Arg173. However, upon examination of free binding energy of D2 as well as C3 first poses with docking scores of −3.7 showed a positive free binding energy, thus they were excluded from further analysis with the second pose of **7b** being used for obtaining data. All the four compounds **13a, 13d, 12d,** and **7b** showed interaction with Tyr60A and Trp60D through π-π- stack and/or hydrophobic interactions. Compounds **7b, 12d,** and **13a** formed hydrophobic interactions with His57. Additionally, **12d** and **13d** formed π-π-stack bond with His57. Finally, all the four compounds maintained hydrophobic interactions with Leu99 and/or Trp215.

EGFR active site comprised from the residues Leu-694, Val-702, Tyr-703, Lys-704, Ala-719, Ile-720, Lys-721, Glu-738, Met-742, Cys-751, Leu-753, Leu-764, Ile-765, Thr-766, Leu-768, Met-769, Pro-770, Phe-771, Gly-772, Cys-773, Leu-820, Thr-830 and Asp-831 [65]. Met-769 is an essential residue of EGFR hinge region contributing to protein structure stabilization or functional regulation connecting the N- and C-lobes, making it a key residue for multiple EGFR inhibitors as it interacts with the 4-aniline quinazoline [66,67]. All of selected compounds could maintain H-bonding with such residue along with H-bonding with LYS721 and/or ASP831 (except for compound **7d**); both of which is located in binding pocket and essential for EGFR functionality [68], where LYS721 with Glu738 forms an ion pair interacting with phosphate

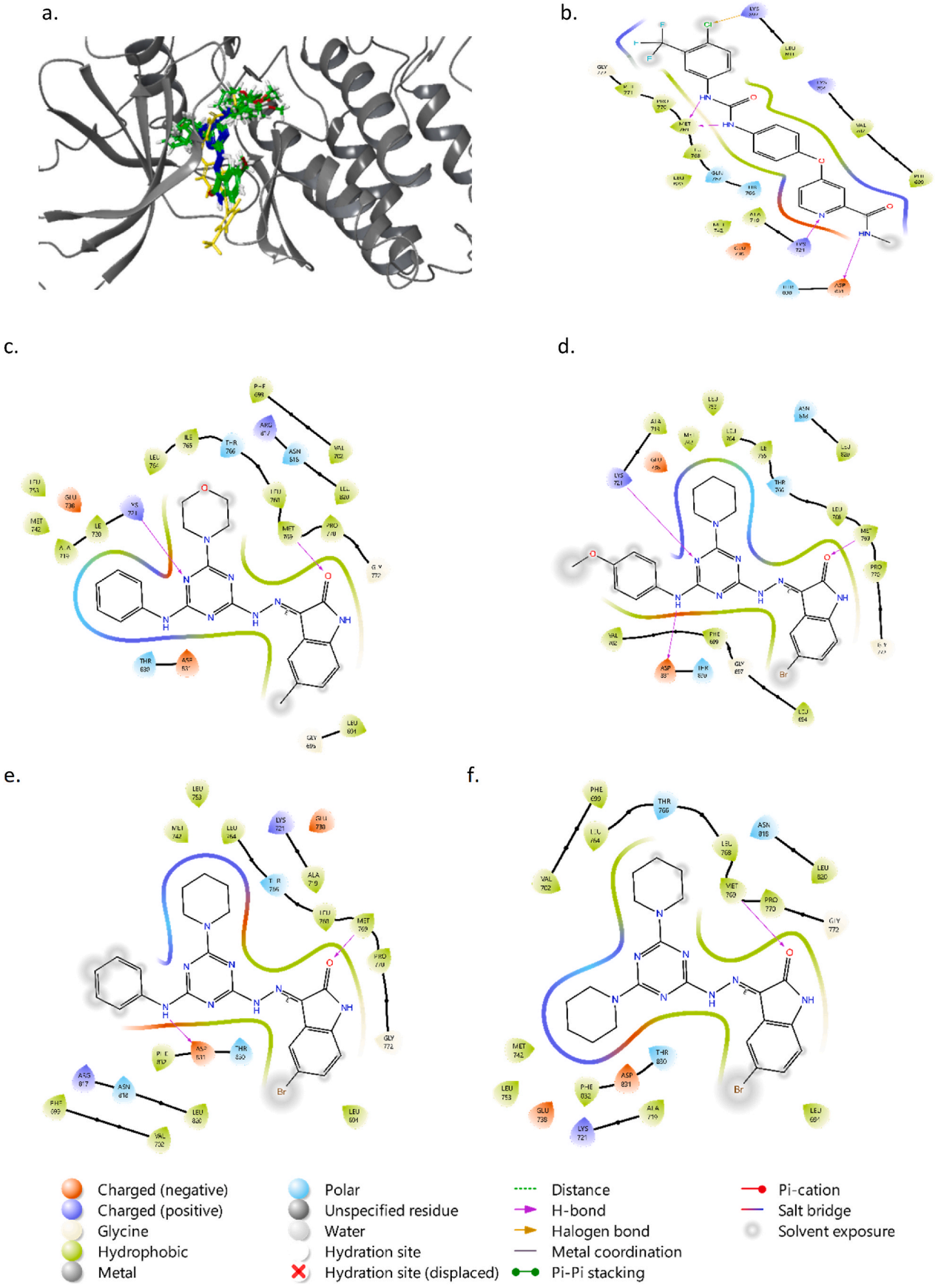


Fig. 5. **a**) Showing Selected compounds (green) superimposed on Sorafenib (Yellow) in the binding Pocket of EGFR (PDB:1M17). b, c, d, e, and f) Showing Interactions of Sorafenib, **11e, 14c, 10c,** and **7c** with the specific residues responsible for binding, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

groups of ATP, thus binding to or deficiency of either residue would result in EGFR initiation inhibition [69,70]. On the other hand, ASP831 is a part of the regulatory activation loop Asp831-Val852 [71]. Additionally, all the selected compounds maintained hydrophobic interactions with LEU694 and LEU820 with some of them achieving more hydrophobic interactions with other residues constituting the active site which demonstrate wide coverage of for the active site.

Factor Xa, a serine protease with the catalytic triad formed by residues His57, Asp102 and ser195. Other than the catalytic triad two pockets namely S1 and S4 pockets were identified as parts of active site and targets for inhibition of Factor Xa [72]. Direct inhibitors binding occurs in L-shaped manner and mainly depends on interactions with S1 and S4 pockets [73]. Upon docking of the selected compounds with 2CJI crystalline structure of Factor Xa all the compounds showed similar mode of binding interacting with both pockets through both hydrophobic and/or ionic interactions occupying them which explains high efficacy demonstrated in invitro testing. Even though **6c** and **6b** lacked the L-shaped binding however having hydrogen bonding with Gly216 and/or Ser214 along with maintaining hydrophobic interactions with all residues of S4 pocket can explain its ability to block the availability of S1 deep pocket maintaining high inhibition activity. For compounds **9b, 9c,** and **13a** having extra H-bonding with Ser195 of catalytic triad and

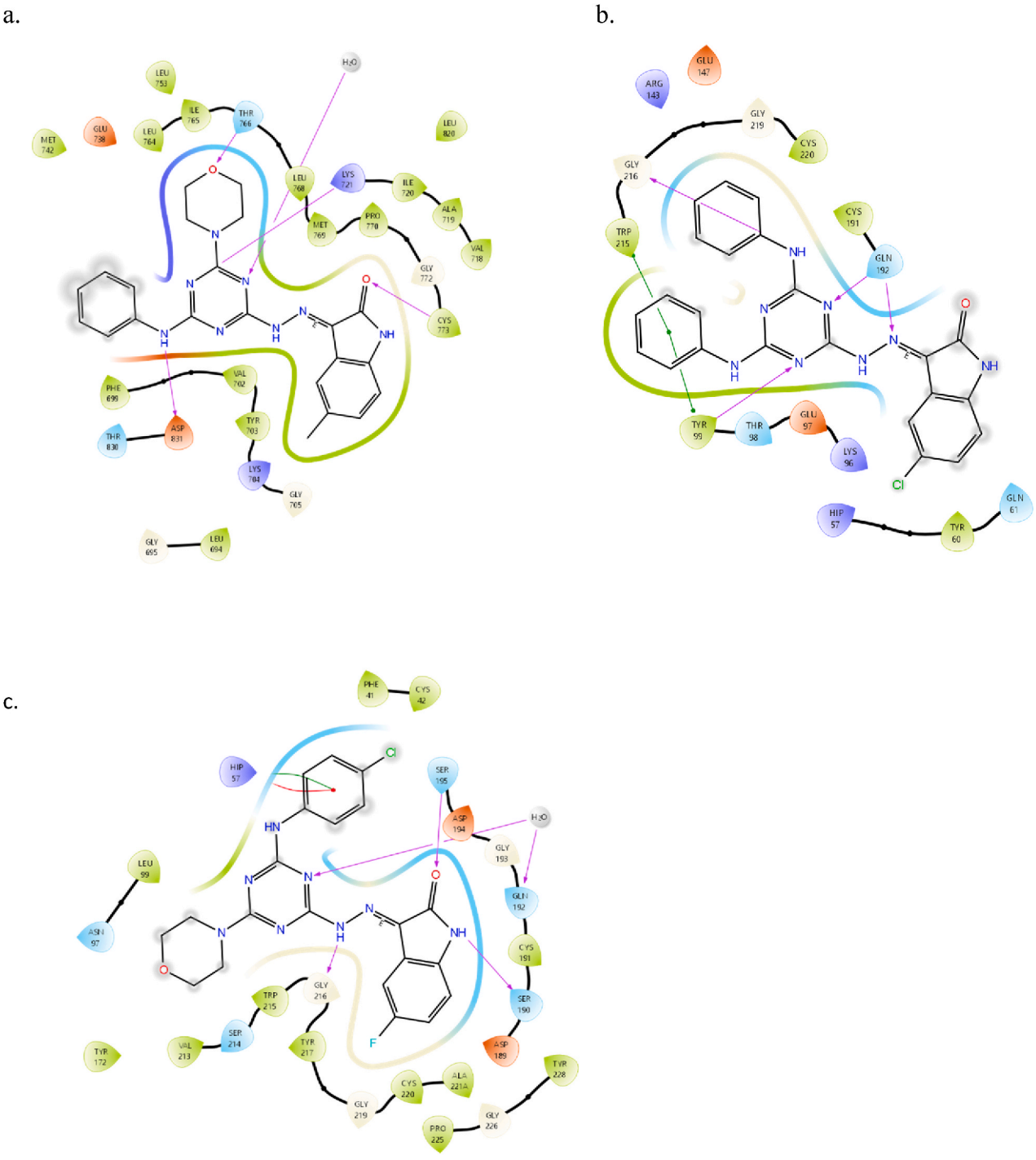


Fig. 6. Showing Induced fit ligand interactions with proteins, a) Shows K6 binding interactions with 1M17, b) Shows E3 binding interactions with 2CJI, and c) Shows G5 binding interactions with 1S85.

Gln61 located near it can suggest their ability to block access of substrates to the triad promoting the inhibitory activity, however the difference seen in activity can be explained through examination of hydrophobic interactions where compound **9c** (most potent) showed interaction with the entire S4 pocket combined with multiple S1 residues including the hinge residue Cys220. Finally, compound **9d** showed a unique extra binding with Arg222, a residue that was identified as a residue of a putative exosite, a main allosteric inhibition target in Factor Xa [74]; rendering it a potential candidate for mixed type inhibition of Factor Xa.

Both trypsin and thrombin are serine proteases showing high similarities where both has a catalytic triad like that of factor Xa as the core of their active site. Trypsin has a main substrate binding pocket formed by residues 189–195, 214–220, and 225–228 known as S1 pocket and an oxyanion hole "Gly193 and Ser195" both of which are located near the catalytic triad [75] All the chosen compounds maintained an interaction with at least one residue of catalytic triad with exception of compound **7b** which formed H-bond with Gly193 of oxyanion hole. Such interactions along with multiple hydrophobic interactions with S1 pocket residues demonstrate a proper potential explanation of high biological activity of these compounds against Trypsin. On the hand, regarding thrombin PDB:1KTS only four compounds **13a, 12d, 13d** and **7b** were considered for analysis. Arg173 of thrombin Exosite II was identified to be an essential residue in allosteric inhibition of thrombin [76,77]. Both compounds **13a** and **12d** form H-bonding with allosteric inhibition target residue Arg173, while maintaining one or more interactions with His57 a member if the catalytic triad and a target of active site inhibitors. Both compounds **13d** and **7b** also showed interactions with His57. Finally, all compounds showed interaction with Tyr60A and Trp60D members of 60-Loop, which forms rim of the active site [78], as well as Hydrophobic interaction with Leu99 and/or Trp215, both of which are essential residues in aryl binding site of thrombin which is involved in substrate recognition [79] Combining all interaction suggesting that both compounds **13d** and **7b** can be considered direct active site inhibitors of thrombin. On the other hand, compounds **13a** and **12d** can be considered mixed type inhibitors combining allosteric inhibition with active site inhibition.

2.2.3. *Induced fit*

Upon docking of compounds with the highest *in vitro* activity with the previously prepared protein structures, extra potential interactions was revealed allowing better in-depth, determination of potential binding and mechanism of action (Fig. 6, Table S4, *Supplementary Data*). Compound **11e** (the most potent tested EGFR inhibitor) maintained h-bonding with Lys721 and showed additional h-bonding with ASP831, Thr766 and Cys773. While compound **9b** (the most potent tested Factor Xa inhibitor) showed high interaction with factor Xa S4 pocket with pi-pi stack bonding with Trp215 and Tyr99 and H-bonding with Tyr99, along with maintaining H-bonding with S1 pocket residues Gln192 and Gly216. Finally, **12d** (the most potent tested Factor Xa inhibitor) was docked with Trypsin Structure (PDB:1S85) showing it maintained interactions with the catalytic triad through H-bond with Ser195 and dual interaction with His57 through π-π stack and π-cation bonding, with h-bonding with Gln192 (through water bridge) and additional ones with Ser19 and Gly216.

Thus, compound **11e** kept interaction with EGFR active site residues including Lys721 and Asp831 associated with initiation and activation of EGFR respectively, thus inhibiting its activity. Compound **9b** showed high interaction with Factor Xa S4 pocket residues Tyr99 and Trp215 essential for substrate binding along with maintaining of L-shaped binding through bindings and hydrophobic interactions with S1 and S4 pocket residues, thus blocking substrate binding and activity. Finally, **12d** showed binding with Trypsin active site catalytic triad through interactions with His 57 and Ser195 along with binding with residue Gln192 located near the oxyanion hole thus diminishing the catalytic triad activity.

3. Conclusion

This study involved the synthesis and characterization of fifty new *s*-triazine-isatin hydrazone derivatives. The compounds were tested for antiproliferative activity against A549 lung cancer cells and WI-38 normal lung cells. The results showed that all compounds had lower IC_{50} values against A549 cells compared to WI-38 cells, indicating their relative safety. Many compounds exhibited safety indices equal to or higher than sorafenib's index of 11.025, with several showing particularly promising profiles.

Regarding selectivity, some compounds demonstrated superior potency compared to sorafenib, with selectivity indices greater than 1. The compounds also showed broad-spectrum anti-trypsin activity, with several surpassing the inhibitory effect of Bis(5-amidino-2-benzimidazolyl)methane. For Factor Xa inhibition, most compounds had lower inhibitory effects compared to rivaroxaban, except for a few that exhibited significant inhibition.

Additionally, all compounds acted as epidermal growth factor (EGF) inhibitors to varying extents. Several compounds showed greater EGF inhibition compared to sorafenib, with improved inhibitory effects at their IC_{50} values. Docking studies confirmed that these compounds had higher docking scores and more favorable binding energies than sorafenib.

In general, most of the compounds contain anilino substituent and cyclic aliphatic amine attached to the triazine moiety (**GP V-X**) demonstrated superior EGF inhibition compared to their analogs *bis*-anilino (**PGIV**) and cyclic aliphatic amine (**GP I-III**). Especially the group incorporates the anilino derivatives and morpholino moiety attached to the triazine ring (**GPX**) showed better inhibition performance compared to Sorafenib as standard drug used.

The work described in this manuscript has demonstrated that the triazine core is a privileged structure for drug development. However, the type of substituent on the triazine ring has also an important impact on the biological activity. Furthermore, we have demonstrated that the integration of two privileged structures, triazine and isatin in this case, in just a single structure have synergic effects.

4. Experimental section

4.1. *Chemistry*

4.1.1. *Materials and methods*

The solvents used were of analytical reagent grade. ^{1}H- and ^{13}C-NMR spectra were recorded on a JEOL 500 MHz spectrometer at room temperature. The chemical shifts were measured using internal standard $\delta = 0$ ppm (TMS). Elemental analyses were performed on PerkinElmer 2400 elemental analyzer. Melting points were recorded on a Mel-Temp apparatus in an open capillary and are uncorrected. Fourier transform infrared spectroscopy (FTIR) spectra were recorded on Nicolet 560 spectrophotometer from KBr discs. The reaction was followed-up and checks for the purity were detected using TLC on silica gel-protected aluminum sheets (Type 60 GF254, Merck).

4.1.2. Synthetic methodology synthesis of the compounds 2-4a-j was performed using the reported method [57] and their spectral characterizations are agreed with the reported data (*Method S2, Supplementary Data*).

4.1.2. *General method for the synthesis of s-triazine hydrazone derivatives GPI-GPX (6-15a-e)*

The synthesis of the target was preformed using the reported method [56]: isatin derivatives **5a-e** (2 mmol) was dissolved in 30 mL ethanol in the presence of 2–3 drops of acetic acid (AcOH). Then *s*-triazine hydrazine derivatives **4a-j** (2 mmol) was added portion wise on the hot solution under stirred. After complete addition, the reaction mixture was refluxed for 6–8 h (the reaction followed by TLC using ethylacetate-hexane; 2:1). At the end, the reaction was left to cool slowly

to room temperature and then the product was collected by filtration, washed with cold ethanol, and dried to produce the pure products, as indicated from their spectral data and Elemental analysis.

4.1.2.1. 3-(2-(4,6-Dimorpholino-1,3,5-triazin-2-yl)hydrazineylidene) indolin-2-one (6a). The product was obtained as a yellow solid in 90 % yield; m.p. 324–326 °C; IR (KBr, cm^{-1}): 3201 (NH), 1581 (C═N), 15496, 1442 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.61–3.63 (m, 8H, 4 CH_2–O), 3.73 (brs, 8H, 4 CH_2–N), 6.92(d, 1H, J = 8Hz, CH), 7.05–7.08 (m, 1H, CH), 7.23–7.33 (m, 1H, CH), 7.51(d, 1H, J = 7.5 Hz, CH), 11.16 (s, 1H, NH–N =), 12.47 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.3, 65.9, 110.8, 119.9, 122.3, 130.0, 132.0, 136.0, 141.2, 163.0, 163.9, 164.6 ppm. Anal. for $C_{19}H_{22}N_8O_3$ (410.44) Calc: C, 55.60; H, 5.40; N, 27.30. Found: C, 55.43; H, 5.21; N, 27.43.

4.1.2.2. 5-Bromo-3-(2-(4,6-dimorpholino-1,3,5-triazin-2-yl)hydrazineylidene)indolin-2-one (6b). The product was obtained as a yellow solid in 88 % yield; m.p. 339–340 °C; IR (KBr, cm^{-1}): 3209 (NH), 1581 (C═N), 1496, 1350 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.63 (brs, 8H, 4 CH_2–O), 3.74 (brs, 8H, 4 CH_2–N), 6.91(d, 1H, J = 5.0 Hz, CH), 7.49(d, 1H, J = 5.0 Hz, CH), 7.65 (s, 1H, CH), 11.30 (s, 1H, NH–N =), 12.46 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.8, 66.5, 113.4, 114.6, 122.7, 123.0, 131.9, 132.9, 141.0, 163.2, 164.2, 165.1 ppm. Anal. for $C_{19}H_{21}BrN_8O_3$ (489.33) Calc: C, 46.64; H, 4.33; N, 22.90. Found: C, 46.53; H, 4.22; N, 23.03.

4.1.2.3. 5-Chloro-3-(2-(4,6-dimorpholino-1,3,5-triazin-2-yl)hydrazineylidene)indolin-2-one (6c). The product was obtained as a yellow solid in 89 % yield; m.p. 343–345 °C; IR (KBr, cm^{-1}): 3170 (NH), 1581 (C═N), 1496, 1450 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.59–3.60 (m, 8H, 4 CH_2–O), 3.69 (brs, 8H, 4 CH_2–N), 6.90 (d, 1H J = 8.5 Hz, CH), 7.31 (dd, 1H, J^4 = 2.5, J^3 = 8.5 Hz, CH), 7.46 (d, 1H, J = 2.5 Hz, CH), 11.26 (s, 1H, NH–N =), 12.42 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.4, 65.9, 112.0, 119.4, 122.2, 126.5, 129.6, 131.0, 139.0, 162.8, 163.8, 164.6 ppm. Anal. for $C_{19}H_{21}ClN_8O_3$ (444.88) Calc: C, 51.30; H, 7.97; N, 25.19. Found: C, 51.18; H, 7.79; N, 25.01.

4.1.2.4. 1-((4,6-Dimorpholino-1,3,5-triazin-2-yl)amino)-6-fluoro-1,3-dihydro-2H-benzo[d]imidazole-2-one (6d). The product was obtained as a yellow solid in 94 % yield; m.p. 319–320 °C; IR (KBr, cm^{-1}): 3186 (NH), 1581 (C═N), 1496, 1442 (C═C);^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.62–3.64 (m, 8H, 4 CH_2–O), 3.74 (brs, 8H, 4 CH_2–N), 6.93–6.95 (m, 2H, 2CH), 7.16(td, 1H, J^4 = 5.0, J^3 = 10.0 Hz, CH), 7.36(dd, 1H, J^4 = 5.0, J^3 = 10.0 Hz, CH), 11.21 (s, 1H, NH–N =), 12.48 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.8, 66.5, 107.6, 112.4, 117.2, 122.1, 132.6, 137.9, 157.8, 163.6, 164.2, 165.1 ppm. Anal. for $C_{19}H_{21}FN_8O_3$(428.43) Calc: C, 53.27; H, 4.94; N, 26.16. Found: C, 53.41; H, 5.02; N, 26.01.

4.1.2.5. 3-(2-(4,6-Dimorpholino-1,3,5-triazin-2-yl)hydrazineylidene)-5-methylindolin-2-one(6e). The product was obtained as a yellow solid in 82 % yield; m.p. 333–335 °C; IR (KBr, cm^{-1}): 3209 (NH), 1581 (C═N), 1519, 1357 (C═C);^{1}H NMR (500 MHz, DMSO-d_6) δ = 2.26 (s, 3H, CH_3), 3.58–3.69 (m, 8H, 4 CH_2–O), 3.96 (brs, 8H, 4 CH_2–N), 6.77 (d, 1H, J = 8.0 Hz, CH), 7.09 (d, 1H, J = 2.5 Hz, CH), 7.30 (s, 1H, CH), 11.06 (s, 1H, NH–N =), 12.48 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 20.6, 43.3, 66.0, 110.6, 120.3, 120.4, 130.8, 131.4, 132.7, 139.5, 163.1, 163.8, 164.7 ppm. Anal. for $C_{20}H_{24}N_8O_3$ (424.47) Calc: C, 56.59; H, 5.70; N, 26.40. Found: C, 56.38; H, 5.59; N, 26.60.

4.1.2.6. 3-(2-(4,6-Di(piperidin-1-yl)-1,3,5-triazin-2-yl)hydrazineylidene) indolin-2-one (7a). The product was obtained as a yellow solid in 82 % yield; m.p. 308–312 °C; IR (KBr, cm^{-1}): 3209 (NH), 1581 (C═N), 1496, 1442 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.47–1.48 (m, 8H, 4CH_2), 1.57–1.58 (m, 4H, 2CH_2), 3.69 (brs, 8H, 4 CH_2–N), 6.89 (d, 1H J = 8.0 Hz, CH), 7.03–7.04 (m, 1H, CH), 7.26–7.29 (m, 1H, CH), 7.48 (d, 1H, J = 7.5 Hz, CH), 11.13 (s, 1H, NH–N =), 12.39 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.3, 25.4, 43.6, 110.8, 119.8, 120.5, 122.3, 130.0, 131.9, 141.0, 162.9, 163.9, 164.4 ppm. Anal. for $C_{21}H_{26}N_8O$ (406.49) Calc: C, 62.05; H, 6.45; N, 27.57. Found: C, 61.98; H, 6.37; N, 27.73.

4.1.2.7. 5-Bromo-3-(2-(4,6-di(piperidin-1-yl)-1,3,5-triazin-2-yl)hydrazineylidene)indolin-2-one(7b). The product was obtained as a yellow solid in 89 % yield; m.p. 326–327 °C; IR (KBr, cm^{-1}): 2931 (NH), 1589 (C═N), 1496, 1350 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.46–1.47 (m, 8H, 4CH_2), 1.56–1.57 (m, 4H, 2CH_2), 3.68 (brs, 8H, 4CH_2–N), 6.84 (d, 1H, J = 8.5 Hz, CH), 7.42 (dd, 1H, J^4 = 2.0, J^3 = 8.25 Hz, CH), 7.56 (d, 1H, J = 1.5 Hz, CH), 11.25 (s, 1H, NH–N =), 12.37 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.2, 25.4, 43.6, 112.7, 114.0, 121.9, 122.6, 130.8, 132.2, 140.0, 162.7, 163.7, 164.3 ppm. Anal. for $C_{21}H_{25}BrN_8O$ (485.39) Calc: C, 51.96; H, 5.19; N, 23.09. Found: C, 51.86; H, 5.27; N, 23.21.

4.1.2.8. 5-Chloro-3-(2-(4,6-di(piperidin-1-yl)-1,3,5-triazin-2-yl)hydrazineylidene)indolin-2-one (7c). The product was obtained as a yellow solid in 92 % yield; m.p. 327–331 °C; IR (KBr, cm^{-1}): 3263 (NH), 1597 (C═N), 1496, 1350 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.47–1.48 (m, 8H, 4CH_2), 1.57–1.58 (m, 4H, 2CH_2), 3.69 (brs, 8H, 4CH_2–N), 6.90 (d, 1H, J = 8.5 Hz, CH), 7.30 (dd, 1H, J^4 = 2.0, J^3 = 8.25 Hz, CH), 7.45 (d, 1H, J = 2.0 Hz, CH), 11.25 (s, 1H, NH–N =), 12.38 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.3, 25.4, 43.6, 112.3, 119.3, 122.3, 126.4, 129.4, 130.9, 139.7, 162.9, 163.8, 164.3 ppm. Anal. for $C_{21}H_{25}ClN_8O$ (440.94) Calc: C, 57.20; H, 5.72; N, 25.41. Found: C, 57.39; H, 5.84; N, 25.25.

4.1.2.9. 3-(2-(4,6-Di(piperidin-1-yl)-1,3,5-triazin-2-yl)hydrazineylidene)-5-fluoroindolin-2-one(7d). The product was obtained as a yellow solid in 77 % yield; m.p. 304–306 °C; IR (KBr, cm^{-1}): 3170 (NH), 1589 (C═N), 1496, 1350 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.31–1.31 (m, 8H, 4CH_2), 1.41–1.42 (m, 4H, 2CH_2), 3.53 (brs, 8H, 4CH_2–N), 6.70–6.73 (m, 1H, CH), 6.92–6.96 (m, 1H, CH), 7.10–7.13 (m, 1H, CH), 10.97 (s, 1H, NH–N =), 12.24 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.2, 25.4, 43.6, 106.9, 111.8, 116.4, 121.8, 131.5, 137.2, 157.3, 163.1, 163.8, 164.3 ppm. Anal. for $C_{21}H_{25}FN_8O$ (424.48) Calc: C, 59.42; H, 5.94; N, 26.40. Found: C, 59.22; H, 5.81; N, 26.59.

4.1.2.10. 3-(2-(4,6-Di(piperidin-1-yl)-1,3,5-triazin-2-yl)hydrazineylidene)-5-methylindolin-2-one(7e). The product was obtained as a yellow solid in 89 % yield; m.p. 280–281 °C; IR (KBr, cm−1): 3209 (NH), 1573 (C═N), 1504, 1442 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.33–1.34 (m, 8H, 4CH_2), 1.44–1.45 (m, 4H, 2CH_2), 2.13 (s, 3H, CH_3), 3.55 (brs, 8H, 4CH_2–N), 6.64 (d, 1H, J = 8.0 Hz, CH), 6.93–6.95 (m, 1H, CH), 7.17 (s, 1H, CH), 10.88 (s, 1H, NH–N =), 12.29 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 20.6, 24.3, 25.4, 43.6, 110.5, 120.2, 120.5, 130.5, 131.3, 132.0, 139.0, 163.1, 163.9, 164.4 ppm. Anal. for $C_{22}H_{28}N_8O$ (420.52) Calc: C, 62.84; H, 6.71; N, 26.65. Found: C, 62.74; H, 6.62; N, 26.53.

4.1.2.11. 3-(2-(4-Morpholino-6-(piperidin-1-yl)-1,3,5-triazin-2-yl)hydrazineylidene)indolin-2-one(8a). The product was obtained as a yellow solid in 71 % yield; m.p. 305–408 °C; IR (KBr, cm^{-1}): 3140 (NH), 1581 (C═N), 1496, 1350; ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.48 (m, 4H, 2CH_2), 1.57–1.58 (m, 2H, CH_2), 3.58–3.60 (m, 4H, 2CH_2–O), 3.69 (brs, 8H, 4CH_2–N), 6.89 (d, 1H, J = 8.0 Hz, CH), 7.01–7.04 (m, 1H, CH), 7.26–7.29 (m, 1H, CH), 7.48 (d, 1H, J = 7.5 Hz, CH), 11.13 (s, 1H, NH–N =), 12.40 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.2, 25.4, 43.3, 43.6, 66.5, 111.9, 120.5, 121.5, 123.0, 130.9, 133.0, 141.9, 163.8, 164, 165, 165.8 ppm. Anal. for $C_{20}H_{24}N_8O_2$ (408.47) Calc: C, 58.81; H, 5.92; N, 27.43. Found: C, 59.02; H, 6.03; N, 27.22.

4.1.2.12. 5-Bromo-3-(2-(4-morpholino-6-(piperidin-1-yl)-1,3,5-triazin-2-yl)hydrazineylidene)indolin-2-one (8b). The product was obtained as a yellow solid in 75 % yield; m.p. 346–350 °C; IR (KBr, cm^{-1}): 3201 (NH), 1581 (C═N), 1496, 1357; ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.47 (m, 4H, $2CH_2$), 1.57–1.58 (m, 2H, CH_2), 3.58–3.60 (m, 4H, $2CH_2$–O), 3.69 (brs, 8H, $4CH_2$–N), 6.85 (d, 1H, J = 8.0 Hz, CH), 7.43-7-7.45 (m, 1H, CH), 7.57 (d, 1H, J = 2.0 Hz, CH), 11.27 (s, 1H, NH–N =), 12.40 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.2, 25.4, 43.3, 43.6, 66.0, 112.8, 114.1, 122.1, 122.6, 131.1, 132.3, 140.2, 162.7, 163.7, 164.2 ppm. Anal. for $C_{20}H_{23}BrN_8O_2$ (487.36) Calc: C, 49.29; H, 4.76; N, 22.99. Found: C, 49.42; H, 4.91; N, 23.12.

4.1.2.13. 5-Chloro-3-(2-(4-morpholino-6-(piperidin-1-yl)-1,3,5-triazin-2-yl)hydrazineylidene)indolin-2-one(8c). The product was obtained as a yellow solid in 79 % yield; m.p. 322–325 °C; IR (KBr, cm^{-1}): 3186 (NH), 1581 (C═N), 1489, 1357; ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.46 (s, 4H, $2CH_2$), 1.57–1.58 (m, 2H, CH_2), 3.58–3.60 (m, 4H, $2CH_2$–O), 3.68 (brs, 8H, $4CH_2$–N), 6.89 (d, 1H, J = 8.5 Hz, CH), 7.29-7-7.31 (m, 1H, CH), 7.45 (d, 1H, J = 2.0 Hz, CH), 11.25 (s, 1H, NH–N =), 12.37 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.2, 25.4, 43.3, 43.6, 66.0, 112.3, 119.3, 122.2, 126.5, 129.5, 131.2, 139.7, 162.8, 163.7, 164.2, 164.8 ppm. Anal. for $C_{20}H_{23}ClN_8O_2$ (442.91) Calc: C, 54.24; H, 8.00; N, 25.30. Found: C, 54.43; H, 8.14; N, 25.51.

4.1.2.14. 5-Fluoro-3-(2-(4-morpholino-6-(piperidin-1-yl)-1,3,5-triazin-2-yl)hydrazineylidene) indolin-2-one (8 d). The product was obtained as a yellow solid in 75 % yield; m.p. 306–310 °C; IR (KBr, cm^{-1}): 3194 (NH), 1589 (C═N), 1496, 1357; ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.51 (s, 4H, $2CH_2$), 1.61–1.62 (m, 2H, CH_2), 3.62–3.64 (m, 4H, $2CH_2$–O), 3.73 (brs, 8H, $4CH_2$–N), 6.92–6.95 (m, 1H, CH), 7.16(td, 1H, J^4 = 5.0, J^3 = 10.0 Hz, CH), 7.36(dd, 1H, J^4 = 5.0, J^3 = 10.0 Hz, CH), 11.20 (s, 1H, NH–N =), 12.46 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.7, 25.9, 43.3, 44.1, 66.5, 107.4, 112.3, 117.1, 122.2, 132.3, 137.8, 157.8, 163.6, 164.2, 165.2 ppm. Anal. for $C_{20}H_{23}FN_8O_2$ (426.46) Calc: C, 56.33; H, 5.44; N, 26.28. Found: C, 56.52; H, 5.59; N, 26.08.

4.1.2.15. 5-Methyl-3-(2-(4-morpholino-6-(piperidin-1-yl)-1,3,5-triazin-2-yl)hydrazineylidene) indolin-2-one (8e). The product was obtained as a yellow solid in 61 % yield; m.p. 325–328 °C; IR (KBr, cm^{-1}): 3209 (NH), 1573 (C═N), 1504, 1442; ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.47 (s, 4H, $2CH_2$), 1.57–1.58 (m, 2H, CH_2), 2.32 (s, 3H, CH_3), 3.58–3.60 (m, 4H, $2CH_2$–O), 3.68–3.81 (m, 8H, $4CH_2$–N), 6.77 (d, 1H, J = 8.0 Hz, CH), 7.07 (d, 1H, J = 7.0 Hz, CH), 7.30 (s, 1H, CH), 11.03 (s, 1H, NH–N =), 12.45 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 20.6, 24.2, 25.4, 43.3, 43.6, 65.9, 110.6, 120.3, 120.5, 130.6, 131.3, 132.4, 138.9, 163.1, 163.9, 164.2, 164.8 ppm. Anal. for $C_{21}H_{26}N_8O_2$ (422.49) Calc: C, 59.70; H, 6.20; N, 26.52. Found: C, 59.91; H, 6.38; N, 26.71.

4.1.2.16. 3-(2-(4,6-Bis(phenylamino)-1,3,5-triazin-2-yl)hydrazineylidene)indolin-2-one(9a). The product was obtained as a yellow solid in 65 % yield; m.p. 279–280 °C; IR (KBr, cm^{-1}): 3186 (NH), 1573 (C═N), 1504, 1442 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 6.98 (d, J = 10.0 Hz,1H, CH), 7.04 (t, J = 7.5 Hz, 2H, 2CH), 7.14 (t, J = 7.5 Hz, 1H, CH), 7.31–7.38 (m, 5H, 5CH), 7.52 (d, J = 10.0 Hz, 1H, CH), 7.78 (brs, 4H, 4CH), 9.79 (s, 2H, 2NH), 11.22 (s, 1H, NH–N =), 12,74 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 110.9, 119.8, 120.3, 120.5, 122.5, 122.6, 128.4, 130.5, 133.3, 139.4, 141.4, 163.1, 163.8, 164.4 ppm. Anal. for $C_{23}H_{18}N_8O$ (422.45) Calc: C, 65.39; H, 4.29; N, 26.53. Found: C, 65.55; H, 4.41; N, 26.41.

4.1.2.17. 3-(2-(4,6-Bis(phenylamino)-1,3,5-triazin-2-yl)hydrazineylidene)-5-bromoindolin-2-one (9b). The product was obtained as a yellow solid in 73 % yield; m.p. 312-213 °C; IR (KBr, cm^{-1}): 3170 (NH), 1581 (C═N), 1504,1442 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 6.74 (d, J = 8.0Hz, 1H, CH), 6.83–6.86 (m, 2H, 2CH), 7.10–7.14 (m, 4H, 4CH), 7.31–7.35 (m,2H, 2CH), 7.59 (s, 4H, 4CH), 9.63 (s, 2H, 2NH), 11.12 (s, 1H, NH–N =), 12.43 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 113.0, 114.0, 120.5, 120.6, 121.9, 122.6, 128.4, 132.0, 132.5, 139.2, 140.5, 162.7, 163.7, 164.4 ppm. Anal. for $C_{23}H_{17}BrN_8O$ (501.35) Calc: C, 55.10; H, 3.42; N, 22.35. Found: C, 55.31; H, 3.53; N, 22.57.

4.1.2.18. 3-(2-(4,6-Bis(phenylamino)-1,3,5-triazin-2-yl)hydrazineylidene)-5-chloroindolin-2-one (9c). The product was obtained as a yellow solid in 68 % yield; m.p. 307–308 °C; IR (KBr, cm^{-1}): 3170 (NH), 1573 (C═N), 1496,1442 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 6.80 (d, J = 8Hz, 1H, CH), 6.84–6.87 (m, 2H, 2CH), 7.12–7.15 (m, 4H, 4CH), 7.20–7.23 (m,2H, 2CH), 7.60 (s, 4H, 4CH), 9.64 (s, 2H, 2NH), 11.13 (s, 1H, NH–N =), 12.47 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 112.6, 119.1, 120.5, 120.6, 122.1, 122.6, 126.5, 128.4, 129.8, 139.0, 140.0, 163.0, 163.7, 164.4 ppm. Anal. for $C_{23}H_{17}ClN_8O$ (456.89) Calc: C, 60.46; H, 3.75; N, 24.53. Found: C, 60.27; H, 3.59; N, 24.65.

4.1.2.19. 3-(2-(4,6-Bis(phenylamino)-1,3,5-triazin-2-yl)hydrazineylidene)-5-fluoroindolin-2-one(9d). The product was obtained as a yellow solid in 61 % yield; m.p. 312-213 °C; IR (KBr, cm^{-1}): 3178 (NH), 1573 (C═N), 1489, 1442 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 6.78–6.80 (m, 1H, CH), 6.84–6.99 (m, 2H, 2CH), 7.02–7.04 (m, 1H, CH), 7.12–7.15 (m, 5H, 5CH), 7.61 (brs, 4H, 4CH), 9.65 (s, 2H, 2NH), 11.04 (s, 1H, NH–N =), 12.50 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 106.5, 112.1, 116.7, 120.5, 121.5, 122.6, 128.4, 132.8, 137.6, 139.4, 157.3, 163.2, 163.7, 164.4 ppm. Anal. for $C_{23}H_{17}FN_8O$ (440.44) Calc: C, 62.72; H, 3.89; N, 25.44. Found: C, 62.91; H, 3.99; N, 25.63.

4.1.2.20. 3-(2-(4,6-Bis(phenylamino)-1,3,5-triazin-2-yl)hydrazineylidene)-5-methylindolin-2-one(9e). The product was obtained as a yellow solid in 80 % yield; m.p. 298–300 °C; IR (KBr, cm^{-1}): 3170 (NH), 1573 (C═N), 1496,1404 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 2.15 (s, 3H, CH_3) 6.66–6.68 (m, 1H, CH), 7.83–7.86 (m, 2H, 2CH), 6.97 (d, J = 8.0 Hz, 1H, CH), 7.11–7.15 (m, 5H, 5CH), 7.60 (brs, 4H, 4CH), 9.61 (s, 2H, 2NH), 10.90 (s, 1H, NH–N =), 12.50 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 20.7, 110.8, 120.1, 120.4, 122.5, 122.6, 128.4, 130.9, 131.4, 133.4, 139.2, 139.4, 163.8, 163.8, 164.4 ppm. Anal. for $C_{24}H_{20}N_8O$(436.48) Calc: C, 66.04; H, 4.62; N, 25.67. Found: C, 66.26; H, 4.71; N, 25.79.

4.1.2.21. 3-(2-(4-(Phenylamino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (10 a). The product was obtained as a yellow solid in 69 % yield; m.p. 268–272 °C; IR (KBr, cm^{-1}): 3200 (NH), 1582 (C═N), 1495, 1442 (C═C); NMR:^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.51 (s, 4H, $2CH_2$), 1.60–1.61 (m, 2H, CH_2), 3.75 (brs, 4H, $2NCH_2$-), 6.91 (d, J = 8.0 Hz, 1H, CH), 6.94–6.97 (m, 1H, CH), 7.05–7.08 (m, 1H, CH), 7.26–7.32 (m, 3H, 3CH), 7.50 (d, J = 7.5 Hz, 1H, CH), 7.71 (brs, 2H, 2CH), 9.80 (s, 1H, NH), 11.20 (s, 1H, NH–N =), 12.50 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.2, 25.4, 44.0, 110.9, 119.9, 120.0, 120.3, 122.3, 122.5, 128.6, 130.5, 133.3, 139.4, 141.5, 163.0, 163.9 ppm. Anal. for $C_{22}H_{22}N_8$(414.47) Calc: C, 63.75; H, 5.35; N, 27.04. Found: C, 63.91; H, 5.44; N, 27.21.

4.1.2.22. 5-Bromo-3-(2-(4-(phenylamino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene) indolin-2-one (10b). The product was obtained as a yellow solid in 68 % yield; m.p. 173–176 °C; IR (KBr, cm^{-1}): 3186 (NH), 1584 (C═N), 1494, 1443 (C═C);^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.51 (s, 4H, $2CH_2$), 1.60–1.61 (s, 2H, CH_2), 3.75 (brs, 4H, $2CH_2$–N), 6.87 (d, J = 8.0 Hz, 1H, CH), 6.94–6.97 (m, 1H, CH), 7.24–7.27 (m, 2H, 2CH), 7.45–7.47 (m, 1H, CH), 7.53 (s, 1H, CH), 7.70 (brs, 2H, 2CH), 9.63 (s, 1H, NH), 11.30 (s, 1H, NH–N =), 12.47 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.2, 25.4, 43.9, 112.9, 114.1, 119.8, 122.0, 122.1, 122.6, 128.5, 131.6, 132.5, 139.8, 140.3, 162.7, 163.0, 164.0 ppm. Anal. for $C_{22}H_{21}BrN_8O$ (493.37) Calc: C, 53.56; H, 4.29; N, 22.71. Found: C, 53.78; H, 4.37; N, 22.93.

4.1.2.23. 5-Chloro-3-(2-(4-(phenylamino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (10c). The product was obtained as a yellow solid in 72 % yield; m.p. 280–282 °C; IR (KBr, cm^{-1}): 3221 (NH), 1589 (C═N), 15497, 1446 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.51 (s, 4H, $2CH_2$), 1.59–1.60 (s, 2H, CH_2), 3.74 (brs, 4H, $2CH_2$–N), 6.91 (d, J = 8.0 Hz, 1H, CH), 6.93–6.96 (m, 1H, CH), 7.24–7.30 (m, 2H, 2CH), 7.31–7.33 (m, 1H, CH), 7.39 (d, J = 2.0 Hz, 1H, CH), 7.71 (brs, 2H, 2CH), 9.60 (s, 1H, NH), 11.27 (s, 1H, NH–N =), 12.48 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.3, 25.4, 43.9, 112.4, 119.2, 119.8, 122.1, 122.2, 126.5, 128.5, 129.6, 131.6, 139.8, 139.9, 162.9, 163.8, 164.3, 172.1 ppm. Anal. for $C_{22}H_{21}ClN_8O$ (448.92) Calc: C, 58.86; H, 7.90; N, 24.96. Found: C, 59.01; H, 8.02; N, 24.81.

4.1.2.24. 5-Fluoro-3-(2-(4-(phenylamino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one(10d). The product was obtained as a yellow solid in 72 % yield; m.p. 295-261 °C; IR (KBr, cm^{-1}): 3212 (NH), 1588 (C═N), 1495, 1444 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.51 (s, 4H, $2CH_2$), 1.60–1.61 (m, 2H, CH_2), 3.75 (brs, 4H, $2CH_2$–N), 6.89–6,96 (m, 2H, 2CH), 7.11–7.15 (m, 1H, CH), 7.21–7.28 (m, 3H, 3CH), 7.72 (brs, 2H, 2CH), 9.61 (s, 1H, NH), 11.19 (s, 1H, NH–N =), 11.99 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.2, 25.4, 43.9, 106.9, 111.9, 116.7, 119.8, 121.7, 122.1, 128.5, 132.2, 137.5, 139.9, 157.3, 163.2, 163.7, 164.3, 172.1 ppm. Anal. for $C_{22}H_{21}FN_8O$ (432.46) Calc: C, 61.10; H, 4.89; N, 25.91. Found: C, 61.33; H, 5.00; N, 25.67.

4.1.2.25. 5-Methyl-3-(2-(4-(phenylamino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (10 e). The product was obtained as a yellow solid in 71 % yield; m.p. 160–164 °C; IR (KBr, cm^{-1}): 3199 (NH), 1584 (C═N), 1493, 1443 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.51 (s, 4H, $2CH_2$), 1.60–1.61 (m, 2H, CH_2), 2.28 (s, 3H, CH_3) 3.74 (brs, 4H, $2CH_2$), 6.79 (d, J = 7.5 Hz, 1H, CH), 6.93–6.96 (m, 1H, CH), 7.10 (d, J = 8.0 Hz, 1H, CH), 7.24–7.28 (m, 3H, 3CH), 7.72 (brs, 2H, 2CH), 9.59 (s, 1H, NH), 11.06 (s, 1H, NH–N =), 11.94 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 21.1, 24.3, 25.5, 44.0, 110.7, 119.7, 120.5, 122.0, 123.0, 128.5, 130.8, 131.4, 134.0, 139.1, 140.0, 163.2, 163.7, 164.0, 172.1 ppm. Anal. for $C_{23}H_{24}N_8O$ (428.50) Calc: C, 64.47; H, 5.65; N, 26.15 Found: C, 64.68; H, 5.74; N, 26.31.

4.1.2.26. 3-(2-(4-Morpholino-6-(phenylamino)-1,3,5-triazin-2-yl)hydrazineylidene)indolin-2-one (11a). The product was obtained as a yellow solid in 88 % yield; m.p. 300–303 °C; IR (KBr, cm^{-1}): 3204 (NH), 1590 (C═N), 1543, 1446 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.62–3.63 (m, 4H,2 CH_2–O), 3.73 (s, 4H, 2 CH_2–N), 6.89 (d, J = 7.5 Hz, 1H, CH), 6.92–6.95 (m, 1H, CH), 7.04–7.07 (m, 1H, CH), 7.24–7.30 (m, 3H, 3CH), 7.45 (d, J = 7.5 Hz, 1H, CH), 7.70 (brs, 2H, 2CH), 9.70 (s, 1H, NH), 11.17 (s, 1H, NH–N =), 12.58 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.5, 65.9, 110.9, 119.9, 120.2, 120.4, 122.2, 122.5, 128.5, 130.4, 132.9, 139.8, 141.4, 163.1, 163.8, 164.3, 164.8 ppm. Anal. for $C_{21}H_{20}N_8O_2$ (416.45) Calc: C, 60.57; H, 4.84; N, 26.91. Found: C, 60.77; H, 4.95; N, 26.74.

4.1.2.27. 5-Bromo-3-(2-(4-morpholino-6-(phenylamino)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (11b). The product was obtained as a yellow solid in 85 % yield; m.p. 299–301 °C; IR (KBr, cm^{-1}): 3202 (NH), 1591 (C═N), 1542, 1442 (C═C); NMR: ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.63–3.64 (m, 4H, $2CH_2$–O), 3.74 (s, 4H, $2CH_2$–N), 6.87 (d, J = 8.0Hz, 1H, CH), 6.96 (t, J = 7.5Hz, 1H, CH), 7.24–7.27 (m, 2H, 2CH), 7.44–7.46 (m, 1H, CH), 7.53 (s, 1H, CH), 7.70 (brs, 2H, 2CH), 9.69 (s, 1H, NH), 11.29 (s, 1H, NH–N =), 12.50 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.5, 65.9, 112.9, 114.1, 119.9, 122.0, 122.2, 122.6, 128.5, 131.7, 132.5, 139.7, 140.4, 162.7, 163.7, 164.2, 165.0 ppm. Anal. for $C_{21}H_{19}BrN_8O_2$ (495.34) Calc: C, 50.92; H, 3.87; N, 22.62. Found: C, 50.78; H, 3.74; N, 22.79.

4.1.2.28. 5-Chloro-3-(2-(4-morpholino-6-(phenylamino)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (11c). The product was obtained as a yellow solid in 88 % yield; m.p. 290–292 °C; IR (KBr, cm^{-1}): 3204 (NH), 1590 (C═N), 1543, 1446 (C═C); ^{1}H NMR (500 MHz, DMSO- d_6) δ = 3.63–3.64 (m, 4H, $2CH_2$–O), 3.74 (s, 4H, $2CH_2$–N), 6.92 (d, J = 8.5 Hz, 1H, CH), 6.94–6.97 (m, 1H, CH),7.24–7.28 (m, 2H, 2CH), 7.32–7.34 (m, 1H, CH), 7.41(s, 1H, CH), 7.70 (brs, 2H, 2CH), 9.69 (s, 1H, NH), 11.28 (s, 1H, NH–N =), 12.51 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.5, 65.9, 112.5, 119.3, 119.9, 122.1, 122.2, 126.5, 128.5, 129.7, 131.9, 139.7, 139.9, 162.9, 163.7, 164.2, 165.0 ppm. Anal. for $C_{21}H_{19}ClN_8O_2$ (450.89) Calc: C, 55.94; H, 4.25; N, 24.85. Found: C, 56.11; H, 4.32; N, 24.73.

4.1.2.29. 5-Fluoro-3-(2-(4-morpholino-6-(phenylamino)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (11d). The product was obtained as a yellow solid in 83 % yield; m.p. 250–254 °C; IR (KBr, cm^{-1}): 3199 (NH), 1583 (C═N), 1547, 1444; ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.63–3.64 (m, 4H, $2CH_2$–O), 3.74 (s, 4H, $2CH_2$–N), 6.89–6.92 (m, 1H, CH), 6.59 (t, J = 7.5Hz, 1H, CH), 7.14 (t, J = 9.0 Hz, 1H, CH), 7.22–7.28 (m, 3H, 3CH), 7.70 (brs, 2H, 2CH), 9.70 (s, 1H, NH), 11.19 (s, 1H, NH–N =), 11.99 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.5, 65.9, 108.0, 112.0, 118.5, 119.9, 121.9, 122.2, 128.5, 132.5, 137.9, 139.7, 157.5, 163.2, 163.9, 164.3, 172.1 ppm. Anal. for $C_{21}H_{19}FN_8O_2$(434.44) Calc: C, 58.06; H, 4.41; N, 25.79. Found: C, 58.23; H, 4.52; N, 25.65.

4.1.2.30. 5-Methyl-3-(2-(4-morpholino-6-(phenylamino)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (11e). The product was obtained as a yellow solid in 86 % yield; m.p. 277–280 °C; IR (KBr, cm^{-1}): 3199 (NH), 1611 (C═N), 1543, 1443; ^{1}H NMR (500 MHz, DMSO-d_6) δ = 2.27 (s, 3H, CH_3), 3.62–3.63 (m, 4H, $2CH_2$–O), 3.73 (s, 4H, $2CH_2$–N), 6.78 (d, J = 8.5Hz, 1H, CH), 6.93–6.95 (m, 1H, CH), 7.09 (d, J = 8.0 Hz, 1H, CH), 7.24–7.27 (m, 3H, 3CH), 7.70 (brs, 2H, 2CH), 9.67 (s, 1H, NH), 11.06 (s, 1H, NH–N =), 12.57 (s, 1H, NH–CO);^{13}C NMR (126 MHz, DMSO-d_6) δ = 20.7, 43.5, 66.0, 111.0, 119.9, 120.3, 120.5, 122.0, 128.5, 131.0, 131.8, 133.5, 139.0, 163.2, 163.7, 164.0, 172.1 ppm. Anal. for $C_{22}H_{22}N_8O_2$ (430.47) Calc: C, 61.38; H, 5.15; N, 26.03. Found: C, 61.26; H, 5.08; N, 26.24.

4.1.2.31. 3-(2-(4-((4-Chlorophenyl)amino)-6-morpholino-1,3,5-triazin-2-yl) hydrazineylidene) indolin-2-one (12a). The product was obtained as a yellow solid in 92 % yield; m.p. 317–320 °C; IR (KBr, cm^{-1}): 3186 (NH), 1612 (C═N), 1496, 1411 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.62–3.63 (m,4H, $2CH_2$–O), 3.72 (brs,4H, $2CH_2$–N), 6.90 (d, J = 7.5 Hz, 1H, CH), 7.04–7.08 (m, 1H, CH), 7.28–7.33 (m, 3H, 3 CH), 7.47 (d, J = 6.0 Hz,1H, CH), 7.74 (brs, 2H, 2CH) , 9.84 (s, 1H, NH), 11.17 (s, 1H, NH–N =), 12.50 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.5, 65.9, 110.9, 119.9, 120.3, 121.3, 122.5, 125.7, 128.4, 130.5, 133.2, 138.0, 141.4, 163.1, 163.8, 164.2, 170.4 ppm. Anal. for $C_{21}H_{19}ClN_8O_2$ (450.89) Calc: C, 55.94; H, 4.25; N, 24.85. Found: C, 56.04; H, 4.31; N, 24.93.

4.1.2.32. 5-Bromo-3-(2-(4-((4-chlorophenyl)amino)-6-morpholino-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (12b). The product was obtained as a yellow solid in 90 % yield; m.p. 324–328 °C; IR (KBr, cm^{-1}): 3262 (NH), 1543 (C═N), 1496, 1411 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.62–3.63 (m, 4H, $2CH_2$–O), 3.72 (brs, 4H, $2CH_2$–N), 6.86 (d, J = 8.5 Hz, 1H, CH), 7.30 (d, J = 9.0 Hz, 2H, 2CH) 7.44–7.46 (m, 1H, CH), 7.53 (s, 1H, CH), 7.71 (brs, 2H, 2CH), 9.81 (s, 1H, NH), 11.29 (s, 1H, NH–N =), 12.50 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.6, 59.8, 112.9, 114.1, 121.4, 122.1, 122.6, 125.8, 128.4, 131.6, 132.6, 138.9, 140.4, 162.7, 163.7, 164.2 ppm. Anal. for $C_{21}H_{18}BrClN_8O_2$ (529.78) Calc: C, 47.61; H, 3.42; N, 21.15. Found: C, 47.88; H, 3.51; N, 21.22.

4.1.2.33. 5-Chloro-3-(2-(4-((4-chlorophenyl)amino)-6-morpholino-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (12c). The product was obtained as a yellow solid in 92 % yield; m.p. 332–336 °C; IR (KBr, cm^{-1}): 3186 (NH), 1681 (C═N), 1496, 1419 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.48–3.50 (m, 4H, 2CH_2–O), 3.60 (brs, 4H, 2CH_2–N), 6.78 (d, J = 8.5 Hz, 1H, CH), 7.16–7.21 (m, 3H, 3CH), 7.28 (s, 1H, CH), 7.58 (brs, 2H, 2CH), 9.66 (s, 1H, NH), 11.13 (s, 1H, NH–N =), 12.39 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.5, 65.9, 112.0, 119.3, 121.3, 122.2, 125.6, 126.5, 128.3, 130.0, 131.9, 138.5, 139.0, 162.9, 163.7, 164.2, 172.1 ppm. Anal. for $C_{21}H_{18}Cl_2N_8O_2$ (485.33) Calc: C, 51.97; H, 3.74; N, 23.09. Found: C, 51.78; H, 3.85; N, 23.23.

4.1.2.34. 3-(2-(4-((4-Chlorophenyl)amino)-6-morpholino-1,3,5-triazin-2-yl)hydrazineylidene)-5-fluoroindolin-2-one (12d). The product was obtained as a yellow solid in 91 % yield; m.p. 322–325 °C; IR (KBr, cm^{-1}): 3178 (NH), 1681 (C═N), 1489, 1419 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.62–3.63 (m,4H, 2CH_2–O), 3.72 (brs,4H, 2CH_2–N), 6.88–6.91(m,1H, CH), 7.11–7.15 (m,1H, CH), 7.23 (d, J = 6.0 Hz, 2H, 2CH), 7.31 (d, J = 9.0 Hz,1H, CH),7.72 (brs, 2H, 2CH), 9.81 (s, 1H, NH), 11.19 (s, 1H, NH–N =), 12.50 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.6, 65.9, 106.9, 112.0, 117.9, 121.4, 121.7, 125.8, 128.4, 132.7, 137.6, 138.9, 157.4, 163.2, 163.2, 163.3, 172.1 ppm. Anal. for $C_{21}H_{18}ClFN_8O_2$ (468.88) Calc: C, 53.79; H, 3.87; N, 23.90. Found: C, 53.95; H, 3.92; N, 23.71.

4.1.2.35. 3-(2-(4-((4-Chlorophenyl)amino)-6-morpholino-1,3,5-triazin-2-yl)hydrazineylidene)-5-methylindolin-2-one(12e). The product was obtained as a yellow solid in 85 % yield; m.p. 330–335 °C; IR (KBr, cm^{-1}): 3209 (NH), 1543 (C═N), 1489, 1411 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 2.32 (s, 3H, CH_3) 3.66–3.68 (m, 4H, 2CH_2–O), 3.78 (brs, 4H, 2CH_2–N), 6.85 (d, J = 10.0 Hz, 1H, CH), 7.16 (d, J = 5.0 Hz,1H, CH), 7.34–7.37 (m, 3H, 3CH), 7.76 (brs, 2H, 2CH), 9.80 (s, 1H, NH), 11.09 (s, 1H, NH–N =), 12.62 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 21.1, 44.0, 66.4, 111.2, 120.8, 121.9, 122.0, 126.3, 128.9, 131.5, 131.9, 133.8, 139.2, 139.6, 163.6, 164.3, 164.6 ppm. Anal. for $C_{22}H_{21}ClN_8O_2$ (464.91) Calc: C, 56.84; H, 4.55; N, 24.10. Found: C, 57.00; H, 4.62; N, 24.23.

4.1.2.36. 3-(2-(4-((4-Chlorophenyl)amino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (13a). The product was obtained as a white solid in yield 94 %; m.p. 254–256 °C; IR (KBr, cm^{-1}): 3281 (NH), 1581(C═N), 1488, 1412 (C═C); ^{1}H NMR (DMSO-d_6): δ = 1.49 (s, 4H, 2CH_2), 1.58–1.59 (m, 2H, CH_2), 3.72 (brs, 4H, 2CH_2–N), 6.89 (d, J = 7.5 Hz,1H, CH), 7.04–7.07 (m, 1H, CH), 7.27–7.32 (m, 3H, 3CH), 7.47 (d, J = 7.0 Hz,1H, CH), 7.75 (brs, 2H, 2CH), 9.76 (s, 1H, NH), 11.17 (s, 1H, NH–N =), 12.56 (s, 1H, NH–CO) ppm; ^{13}C NMR (126 MHz, DMSO-d_6): δ = 24.2, 25.4, 43.9, 110.9, 119.9, 120.4, 121.2, 122.5, 125.6, 128.4, 130.4, 132.9, 138.9, 141.3, 163.1, 163.9, 164.3 ppm. Anal. for $C_{22}H_{21}ClN_8O$ (448.91) Calc: C, 58.86; H, 4.72; N, 24.96. Found: C, 58.99; H, 4.65; N, 24.87.

4.1.2.37. 5-Bromo-3-(2-(4-((4-Chlorophenyl)amino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (13b). The product was obtained as a yellow solid in yield 92 %; m.p. 290–291 °C; IR (KBr, cm^{-1}): 3151 (NH), 1581(C═N), 1486, 1412 (C═C); ^{1}H NMR (DMSO-d_6): δ = 1.37 (s, 4H, 2CH_2), 1.45–1.46 (m, 2H, CH_2), 3.59 (brs, 4H, 2CH_2–N), 6.72 (d, J = 9.0 Hz,1H, CH), 7.15–7.17 (m, 2H, 2CH), 7.29–7.38 (m, 1H, CH), 7.59 (s, 1H, CH), 7.60 (brs, 2H, 2CH), 9.57 (s, 1H, NH), 11.11 (s, 1H, NH–N =), 12.33 (s, 1H, NH–CO) ppm; ^{13}C NMR (126 MHz, DMSO-d_6): δ = 24.2, 25.4, 43.9, 112.9, 114.0, 121.1, 121.9, 122.6, 125.6, 128.3, 131.6, 132.4, 138.9, 140.3, 162.6, 163.7, 164.2, 170.3 ppm. Anal. for $C_{22}H_{20}BrClN_8O$ (527.80) Calc: C, 50.06; H, 3.82; N, 21.23. Found: C, 50.23; H, 3.95; N, 21.51.

4.1.2.38. 5-Chloro-3-(2-(4-((4-Chlorophenyl)amino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (13c). The product was obtained as a yellow solid in yield 99 %; m.p. 291–293 °C; IR (KBr, cm^{-1}): 3291 (NH), 1582(C═N), 1488, 1418 (C═C); ^{1}H NMR (DMSO-d_6): δ = 1.49 (s, 4H, 2CH_2), 1.58–1.59 (s, 2H, CH_2), 3.71 (brs, 4H, 2CH_2–N), 6.89 (d, J = 7.5 Hz,1H, CH), 7.28–7.31 (m, 3H, 3CH), 7.38 (s, 1H, CH), 7.72 (brs, 2H, 2CH), 9.73 (s, 1H, NH), 11.26 (s, 1H, NH–N =), 12.48 (s, 1H, NH–CO)ppm; ^{13}C NMR (126 MHz, DMSO-d_6): δ = 24.2, 25.4, 43.9, 112.4, 119.3, 121.1, 122.2, 125.6, 126.5, 128.3, 129.7, 131.8, 138.9, 139.9, 162.9, 163.7, 164.2, 172.1 ppm. Anal. for $C_{22}H_{20}Cl_2N_8O$ (483.35) Calc: C, 54.67; H, 4.17; N, 23.18. Found: C, 54.89; H, 4.31; N, 23.36.

4.1.2.39. 3-(2-(4-((4-Chlorophenyl)amino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)-5- fluoroindolin-2-one (13d). The product was obtained as a yellow solid in yield 93 %; m.p. 276–278 °C; IR (KBr, cm^{-1}): 3220 (NH), 1582(C═N), 1486, 1420 (C═C); ^{1}H NMR (DMSO-d_6): δ = 1.49 (brs, 4H, 2CH_2), 1.58–1.59 (brs, 2H, CH_2), 3.72 (brs, 4H, 2CH_2–N), 6.87–6.90 (m, 1H, CH), 7.09–7.14 (m, 1H, CH), 7.21 (d, J = 5.5 Hz, 2H, 2CH), 7.31 (d, J = 8.5 Hz, 1H, CH), 7.73 (brs, 2H, 2CH), 9.74 (s, 1H, NH), 11.18 (s, 1H, NH–N =), 12.57 (s, 1H, NH–CO) ppm; ^{13}C NMR (126 MHz, DMSO-d_6): δ = 24.2, 25.4, 43.9, 106.8, 112.0, 116.6, 121.2, 121.7, 125.6, 128.4, 132.4, 137.6, 138.9, 157.3, 163.2, 163.8, 164.2172.1 ppm. Anal. for $C_{22}H_{20}ClFN_8O$ (466.90) Calc: C, 56.59; H, 4.32; N, 24.00. Found: C, 56.71; H, 4.48; N, 24.23.

4.1.2.40. 3-(2-(4-((4-Chlorophenyl)amino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)-5- methylindolin-2-one (13e). The product was obtained as a yellow solid in yield 77 %; m.p. 296–297 °C; IR (KBr, cm^{-1}): 3287 (NH), 1581 (C═N), 1494, 1422 (C═C); ^{1}H NMR (DMSO-d_6): δ = 1.49 (brs, 4H, 2CH_2), 1.58–1.59 (m, 2H, CH_2), 2.27 (s, 3H, CH_3), 3.72 (brs, 4H, 2CH_2–N), 6.77 (d, J = 7.5 Hz, 1H, CH), 7.08 (d, J = 7.5 Hz, 1H, CH),7.26 (s, 2H, 2CH), 7.30(d, J = 8.5 Hz, 1H, CH), 7.72 (brs, 2H, 2CH), 9.73 (s, 1H, NH), 11.05 (s, 1H, NH–N =), 12.55 (s, 1H, NH–CO) ppm; ^{13}C NMR (126 MHz, DMSO-d_6): δ = 21.1, 24.2, 25.4, 43.9, 111.2, 121.1, 121.9, 122.0, 126.3, 128.4, 131.5, 131.9, 133.8, 139.2, 139.6, 163.6, 164.3, 164.6, 172.1 ppm. Anal. for $C_{23}H_{23}FN_8O$ (446.48) Calc: C, 61.87; H, 5.19; N, 25.10. Found: C, 61.96; H, 5.32; N, 25.35.

4.1.2.41. 3-(2-(4-((4-Methoxyphenyl)amino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (14a). The product was obtained as a yellow solid in 63 % yield; m.p. 170–175 °C; IR (KBr, cm^{-1}): 3222 (NH), 1593 (C═N), 1544, 1435 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.49 (brs, 4H, 2CH_2), 1.59–1.60 (m, 2H, CH_2), 3.69 (s, 3H, OCH_3), 3.72 (s, 4H, 2CH_2–N), 6.85 (d, J = 9.0 Hz, 2H, 2CH), 6.90 (d, J = 7.5 Hz, 1H, CH), 7.01–7.07 (m, 1H, CH), 7.27–7.30 (m, 1H, CH), 7.46 (d, J = 7.5 Hz, 1H, CH), 7.59 (brs, 2H, 2CH), 9.52 (s, 1H, NH), 11.16 (s, 1H, NH–N =), 12.51 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.3, 25.5, 43.8, 55.2, 110.9, 113.7, 119.9, 120.5, 121.3, 122.4, 130.3, 132.5, 132.9, 141.3, 154.6, 163.1, 163.8, 164.3, 172.1 ppm. Anal. for $C_{23}H_{24}N_8O_2$ (444.50) Calc: C, 62.15; H, 5.44; N, 25.21. Found: C, 62.36; H, 5.62; N, 25.03.

4.1.2.42. 5-Bromo-3-(2-(4-((4-methoxyphenyl)amino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (14b). The product was obtained as a yellow solid in 86 % yield; m.p. 265–269 °C; IR (KBr, cm^{-1}): 3188 (NH), 1595 (C═N), 1503, 1423 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.50 (s, 4H, 2CH_2), 1.59–1.60 (m, 2H, CH_2), 3.69 (s, 3H, OCH_3), 3.73 (s, 4H, 2CH_2–N), 6.83–6.88 (m, 3H, 3CH), 7.44–7.46 (m, 1H, CH), 7.52 (s, 1H, CH), 7.59 (brs, 2H, 2CH), 9.49 (s, 1H, NH), 11.27 (s, 1H, NH–N =), 12.45 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.3, 25.4, 43.9, 55.2, 112.9, 113.7, 114.1, 121.2, 121.9, 122.7, 131.3, 132.3, 132.9, 140.2, 154.6, 162.7, 163.0, 163.5164.0 ppm. Anal. for $C_{23}H_{23}BrN_8O_2$ (523.40) Calc: C, 52.78; H, 4.43; N, 21.41. Found: C,

52.91; H, 4.58; N, 21.62.

4.1.2.43. 5-Chloro-3-(2-(4-((4-methoxyphenyl)amino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one(14c). The product was obtained as a yellow solid in 89 % yield; m.p. 180–181 °C; IR (KBr, cm^{-1}): 3145 (NH), 1594 (C═N), 1497, 1437 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.50 (s, 4H, $2CH_2$), 1.59–1.60 (m, 2H, CH_2) 3.69 (s, 3H, OCH_3), 3.73 (s, 4H, $2CH_2$–N), 6.84–6.86 (m, 2H, 2CH), 6.91 (d, J = 8Hz, 1H, CH), 7.31–7.33 (m, 1H, CH), 7.40 (s, 1H, CH), 7,58 (brs, 2H, 2CH), 9.49 (s, 1H, NH), 11.28 (s, 1H, NH–N =), 12.47 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.3, 25.5, 43.9, 55.2, 112.5, 113.7, 119.2, 121.3, 122.3, 126.5, 129.6, 131.4, 132.9, 140.0, 154.6, 162.9, 163.0, 163.5, 172.1 ppm. Anal. for $C_{23}H_{23}ClN_8O_2$(478.94) Calc: C, 57.68; H, 4.84; N, 23.40. Found: C, 57.51; H, 4.76; N, 23.59.

4.1.2.44. 5-Fluoro-3-(2-(4-((4-methoxyphenyl)amino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (14d). The product was obtained as a yellow solid in 97 % yield; m.p. 189–192 °C; IR (KBr, cm^{-1}): 3225 (NH), 1545 (C═N), 1494, 1436 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.50 (s, 4H, $2CH_2$), 1.59–1.60 (m, 2H, CH_2), 3.69 (s, 3H, OCH_3), 3.73 (s, 4H, $2CH_2$–N), 6.85 (d, J = 9.0 Hz, 2H, 2CH), 6.89–6.91 (m, 1H, CH), 7.11–7.15 (m, 1H, CH), 7.22 (d, J = 7.0 Hz, 1H, CH), 7,59 (brs, 2H, 2CH), 9.50 (s, 1H, NH), 11.18 (s, 1H, NH–N =), 12.50 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 24.3, 25.5, 43.8, 55.2, 106.6, 111.9, 113.7, 116.4, 121.4, 121.7, 132.1, 132.9, 137.4, 154.6, 157.3, 163.2, 136.7, 172.1 ppm. Anal. for $C_{23}H_{23}FN_8O_2$(462.49) Calc: C, 59.73; H, 5.01; N, 24.23. Found: C, 59.90; H, 5.12; N, 24.41.

4.1.2.45. 3-(2-(4-((4-Methoxyphenyl)amino)-6-(piperidin-1-yl)-1,3,5-triazin-2-yl) hydrazineylidene)-5-methylindolin-2-one(14e). The product was obtained as a yellow solid in 93 % yield; m.p. 184–188 °C; IR (KBr, cm^{-1}): 3186 (NH), 1589 (C═N), 1504, 1435 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 1.50 (s, 4H, $2CH_2$), 1.59–1.60 (m, 2H, CH_2), 2.27 (s, 3H, CH_3), 3.69 (s, 3H, OCH_3), 3.73 (s, 4H, $2CH_2$–N), 6.78 (d, J = 8.0 Hz, 1H, CH) , 6.84 (d, J = 8.5 Hz, 2H, 2CH), 7.09 (d, J = 7.0Hz, 1H, CH), 7.27 (s, 1H, CH), 7,59 (brs, 2H, 2CH), 9.47 (s, 1H, NH), 11.05 (s, 1H, NH–N =), 12.51 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 21.1, 24.3, 25.5, 43.8, 55.2, 110.7, 113.7, 120.2, 120.5, 121.3, 130.7, 131.4, 132.6, 132.9, 139.0, 154.6, 163.2, 163.8, 164.4, 172.1 ppm. Anal. for $C_{24}H_{26}N_8O_2$ (458.53) Calc: C, 62.87; H, 5.72; N, 24.44. Found: C, 62.81; H, 5.83; N, 24.63.

4.1.2.46. 3-(2-(4-((4-Methoxyphenyl)amino)-6-morpholino-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (15a). The product was obtained as a yellow solid in 73 % yield; m.p. 260–262 °C; IR (KBr, cm^{-1}): 3209 (NH), 1612 (C═N), 1496, 1342 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.61–3.63 (m, 2H, $2OCH_2$-), 3.69 (s, 3H, OCH_3), 3.72 (brs, 4H, $2CH_2$–N), 6.85 (d, J = 8.5 Hz,2H, 2CH), 6.91 (d, J = 8.0 Hz, 1H, CH), 7.06 (t, J = 7.5 Hz, 1H, CH), 7.28–7.31 (m, 1H, CH), 7.47 (d, J = 7.5 Hz, 1H, CH), 7.58 (brs, 2H, 2CH), 9.63 (s, 1H, NH), 11.18 (s, 1H, NH–N =), 12.50 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.5, 55.2, 65.9, 110.9, 113.7, 119.9, 120.5, 121.5, 122.5, 130.0, 132.5, 132.9, 141.4, 154.6, 162.7, 163.0, 163.5164.0 ppm. Anal. for $C_{22}H_{22}N_8O_3$ (446.47) Calc: C, 59.18; H, 4.97; N, 25.10. Found: C, 59.32; H, 5.03; N, 25.00.

4.1.2.47. 5-Bromo-3-(2-(4-((4-methoxyphenyl)amino)-6-morpholino-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (15b). The product was obtained as a yellow solid in 70 % yield; m.p 0.284–286 °C; IR (KBr, cm^{-1}): 3178 (NH), 1589 (C═N), 1489, 1350 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.61–3.62 (m, 2H, $2CH_2$–O), 3.68 (s, 3H, OCH_3), 3.71 (brs, 4H, $2CH_2$– N), 6.83–6.87 (m, 3H, 3CH), 7.43–7.45 (m, 1H, 1CH), 7.52 (s, 1H, CH), 7.56 (brs, 2H, 2CH), 9.55 (s, 1H, NH), 11.26 (s, 1H, NH–N =), 12. 47 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.5, 55.2, 65.9, 112.9, 113.7, 114.1, 122.0, 122.6, 127.9, 128.2, 128.6, 128.9, 132.4, 136.8, 162.0, 163.5, 164.0, 172.0 ppm. Anal. for $C_{22}H_{21}BrN_8O_3$ (525.37) Calc: C, 50.30; H, 4.03; N, 21.33. Found: C, 50.47; H, 4.18; N, 21.20.

4.1.2.48. 5-Chloro-3-(2-(4-((4-methoxyphenyl)amino)-6-morpholino-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (15c). The product was obtained as a yellow solid in 76 % yield; m.p. 290–293 °C; IR (KBr, cm^{-1}): 3172 (NH), 1589 (C═N), 1489, 1350 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.60 (s, 2H, $2OCH_2$-), 3.67 (s, 3H, CH_3–O), 3.70 (brs, 4H, $2CH_2$–N), 6.83 (d, J = 8.5 Hz, 2H, 2CH), 6.90 (d, J = 8.5 Hz, 1H, 1CH), 7.29 (s, 1H, CH), 7.39 (s, 1H, CH), 7.55 (brs, 2H,2CH), 9.55 (s, 1H, NH), 11.25 (s, 1H, NH–N =), 12. 48 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.5, 55.1, 65.9, 112.5, 113.7, 119.3, 122.0, 126.5, 127.9, 128.2, 128.6, 128.9, 129.7, 154.7, 162.0, 163.0, 164.0 ppm. Anal. for $C_{22}H_{21}ClN_8O_3$ (480.91) Calc: C, 54.95; H, 4.40; N, 23.30. Found: C, 54.78; H, 4.32; N, 23.16.

4.1.2.49. 5-Fluoro-3-(2-(4-((4-methoxyphenyl)amino)-6-morpholino-1,3,5-triazin-2-yl) hydrazineylidene)indolin-2-one (15d). The product was obtained as a yellow solid in 71 % yield; m.p. 273–275 °C; IR (KBr, cm^{-1}): 3201 (NH), 1589 (C═N), 1496, 1427 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 3.65–3.67 (m, 2H, $2CH_2$–O), 3.73 (s, 3H, OCH_3), 3.76 (brs, 4H, $2CH_2$–N), 6.89 (d, J = 10.0 Hz, 2H, 2CH), 6.94–6.97 (m, 1H, CH), 7.18 (td, 1H, J^4 = 5.0, J^3 = 10.0 Hz, CH), 7.29–7.31 (m, 1H, CH), 7.60 (brs, 2H, 2CH), 9.52 (s, 1H, NH), 11.22 (s, 1H, NH–N =), 12. 58 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 43.9, 55.6, 66.4, 107.4, 112.5, 114.2, 117.2, 122.0, 122.1, 132.8, 133.0, 137.9, 155.2, 157.8, 163.7, 164.1, 164.5, 165.3 ppm. Anal. for $C_{22}H_{21}FN_8O_3$ (464.46) Calc: C, 56.89; H, 4.56; N, 24.13. Found: C, 57.01; H, 4.68; N, 24.01.

4.1.2.50. 3-(2-(4-((4-Methoxyphenyl)amino)-6-morpholino-1,3,5-triazin-2-yl) hydrazineylidene)-5-methylindolin-2-one (15e). The product was obtained as a yellow solid in 85 % yield; m.p. 315–318 °C; IR (KBr, cm^{-1}): 3209 (NH), 1573 (C═N), 1496, 1350 (C═C); ^{1}H NMR (500 MHz, DMSO-d_6) δ = 2.32 (s, 3H, CH_3), 3.66–3.67 (m, 2H, $2CH_2$–O), 3.73 (s, 3H, OCH_3), 3.76 (brs, 4H, $2CH_2$–N), 6.85 (d, J = 5.0 Hz, 1H, CH), 6.89 (d, J = 10.0 Hz, 2H, 2CH), 7.15 (d, J = 10.0 Hz, 1H, CH), 7.33 (s, 1H, CH), 7.61 (brs, 2H, 2CH), 9.51 (s, 1H, NH), 11.08 (s, 1H, NH–N =), 12. 57 (s, 1H, NH–CO); ^{13}C NMR (126 MHz, DMSO-d_6) δ = 21.1, 43.9, 55.6, 66.4, 111.2, 114.2, 120.7, 120.9, 122.0, 131.3, 131.9, 133.1, 133.4, 139.5, 155.2, 163.6, 164.2, 164.6, 165.4 ppm. Anal. for $C_{23}H_{24}N_8O_3$ (460.50) Calc: C, 59.99; H, 5.25; N, 24.33. Found: C, 59.81; H, 5.17; N, 24.45.

4.2. Biology

4.2.1. Cytotoxicity of different compounds on WI-38 and A549

In 96 well plate, 3000 WI-38 cell/well (ATCC: CCL-25) or A549 (ATCC: CCL-185) suspended in 200 μL Culture media (RPMI containing 10 % fetal bovine and 2 % L-glutamine) were plated and cultured in complete RPMI media for 24 h in CO_2 incubator (37 °C, 5 % CO_2, and 90 % relative humidity). After that, 100 μL of different concentration of each compound was suspended in RPMI medium without fetal bovine serum then added to the plate and then the plate was re-incubated for 48 h in CO_2 incubator (37 °C, 5 % CO_2, and 90 % relative humidity). After 24 h of incubation, a volume of 20 μL of MTT (5 mg/ml in PBS) solution was added to each well and plates were incubated for 3 h in CO_2 incubator to allow the MTT to be reacted. After incubation, the plates were centrifuged at 1650 rpm for 10 min and the medium was discarded. The formazan crystals (MTT byproduct) were re-suspended in 100 μL DMSO and reading was measured at a wavelength of 570 nm using optima spectrophotometer for detecting safe dose, which cause 100 % cell viability [80].

The % viability was calculated as follow: $(A_T\text{-}A_b/A_C\text{-}A_b)$ x 100.

$\mathbf{A_T}$ = mean absorbances of cells treated with different concentration

of each plant extract.

$\mathbf{A_C}$ = mean absorbances of control untreated cells with culture medium only.

$\mathbf{A_b}$ = mean absorbances of cells treated with vehicle of plant extract (RPMI without fetal bovine serum)

The cytotoxicity assay of the compound was expressed as IC_{50} and was calculated by the Graphpad Instat software using the % viability calculated from the serial dilutions of each plant extract.

4.2.2. Effect of A549 IC_{50} on each compound on trypsin, Factor Xa and epidermal growth factor receptor kinase

4.2.2.1. Trypsin activity. A volume of 25 μL of trypsin enzyme (0.3 mg/mL) dissolved in 1 mM HCl solution was placed into a test tube, and then 25 μL chemical compound (IC_{50} on A549) dissolved in DMSO for control, was added to this enzyme solution, pre-incubated for 1 h at 37 °C. To this mixture, 200 μL casein substrate solution (1 % in 0.1 M Tris HCl, pH 7.4) was added, incubated at 37 °C for 15 min. To terminate the reaction, a 250 μL of 10 % TCA solution was added to each test tube. Then the precipitate was removed by centrifugation at 10,000 rpm for 10 min. Then the precipitate was removed by centrifugation at 10,000 rpm for 10 min.

According to Umayaparvathi et al. [81], In microplate, 0.1 mL of supernatant was transferred and neutralized was neutralized with 0.2 ml of 0.4 M sodium carbonate and incubated with 0.05 mL of 1 N Folin Ciocalteu's reagent solution at 40 °C for 20 min. Subsequently the absorbance at 660 nm was measured. The enzyme activity was calculated from the standard curve of L-tyrosine. One unit of activity was defined as the amount of enzyme that liberated 1 μg of tyrosine per milliliter of reaction mixture per minute.

The percent inhibition of trypsin activity was calculated as follows:

Trypsin Inhibitory Activity (%) = (A-B)/(A) x 100.

A is the enzyme activity without an inhibitor, and B is the activity in the presence of an inhibitor.

4.2.2.2. Factor Xa activity. Factor Xa enzyme was preincubated with or without the test compound prior to adding the chromogenic substrate to the reaction. Factor Xa cleaved the substrate and liberated the chromogen *p*-nitroanilid (pNA). pNA is a yellow color chromogen that can be measured at 405 nm. Rivaroxaban was used as reference inhibitor. The procedure was done by using a commercial kit (BPS Bioscience, CA, USA) according to manufacturer instructions.

4.2.2.3. EGFR kinase activity. In this assay that performed by using commercial kit (BPS Bioscience, CA, USA), 6 μL kinase assay buffer was mixed with a solution of 1 μL ATP (500 μM), 1 μL protein tyrosine kinase substrate (poly (Glu: Tyr 4:1)), 17 μL H_2O, 5 μL inhibitor buffer for positive control and blank or 5 μL tested compound (IC_{50} on A549). In Blank, 20 μL kinase buffer was added while 20 μL EGFR (1 ng/μL) was added into test and positive control. The plate is incubated for 40 min at 30C, then 50 μL kinase-Glo Max reagent was added to all wells and the plate was allowed to incubate 15 min at room temperature. The luminescence was measured by using the microplate reader.

4.2.2.4. Statistical analyses. All data are presented as mean ± SD. All analyses were carried out using the SPSS program Version 16.0. The data were analyzed by using *t*-test to evaluate the differences treated group with control. Means were statistically compared using Least Significant Difference (LSD) test at 0.05 significance level. Probability values ($P \leq 0.05$) were statistically significant.

4.2.3. Molecular modeling and docking

Computational modelling, simulations and calculations were performed using Schrödinger suite Maestro 13.5.128 (LLC, New York, NY), using different modules such as LigPrep, Protein Preparation Workflow, Grid Generation, Glide Docking, MM-GBSA and induced fit docking.

4.2.3.1. Protein structures collection, preparation and receptor grid generation. Protein structures for EGFR, Thrombin, Trypsin and Factor Xa (1M17,1KTS, 1S85 & 2CJI respectively) were obtained from protein data bank (PDB). Structures were prepared using Schrödinger suite's module "protein preparation workflow", using the "preparation workflow" tab for preprocess step settings were adjusted as default except for activating both "filling missing loops" and "delete water beyond hets" which was adjusted at 8 A. For both optimization and minimization settings was adjusted as default. Structures were minimized with heavy atom convergence to RMSD 0.3 A using force field OPLS4 and water beyond 5 A from hets were deleted.

Following protein preparation receptor grids were generated using "Receptor grid generation" module, receptor was defined by selection of the co-crystallized ligand with its centroid considered as the center of the enclosing box. All settings were adjusted as default. For covalently bound co-crystalized ligands, covalent bond was deleted before protein preparation using delete bond option from build menu.

4.2.3.2. Ligands obtaining and preparation. Structures were drawn using ChemDraw and saved in mol. extension. They were imported to maestro program and prepared by "LigPrep" module using force field OPLS4. Possible ionization states were generated at pH 7.0 ± 2.0 using Epik along with desalting and generating tautomers. Chiralities were determined from 3D structures and at most 250 stereoisomers were generated per ligand.

4.2.3.3. Molecular docking and binding free energy calculation. Molecular docking was performed through "Ligand docking" module using prepared ligand structures along with generated receptor grids. Docking was performed using Extra Precision protocol. Data were analyzed based on docking score along with specific residues interactions and obtained biological activity. Compounds that showed the best score along with superior biological activity were subjected to calculation of binding free energy using the "MM-GBSA" module. For each compound the pose showing the best docking score was used in data analysis unless otherwise stated.

4.2.3.4. Induced fit docking. For induced fit previously prepared protein structures were used and the centroid of co-crystallized ligand was used as the Box center and "Extra precision" docking option was selected. All other options were kept as default. Only compounds that showed the highest biological activity invitro were subjected to induced fit using "induced fit" module using, for better understanding of potential binding site and action. The pose that showed the highest docking score was used for analysis.

CRediT authorship contribution statement

Hessa H. Al-Rasheed: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Sarah A. Al-Khamis:** Writing – original draft, Methodology, Formal analysis. **Assem Barakat:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition. **Aliaa A. Masoud:** Writing – original draft, Methodology, Formal analysis. **Ahmed A. Sobhy:** Writing – original draft, Methodology, Investigation, Formal analysis. **Doaa A. Ghareeb:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Beatriz G. de la Torre:** Writing – review & editing, Supervision, Investigation. **Fernando Albericio:** Writing – review & editing, Supervision, Investigation. **Ayman El-Faham:** Writing – review & editing, Supervision, Project administration, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.tet.2024.134424.

Data availability

The data that has been used is confidential.

References

[1] R.L. Siegel, K.D. Miller, N.S. Wagle, A. Jemal, CA Cancer J. Clin. 73 (2023) 17–48.
[2] H. Sung, J. Ferlay, R.L. Siegel, M. Laversanne, I. Soerjomataram, A. Jemal, F. Bray, CA Cancer J. Clin. 71 (2021) 209–249.
[3] G. Szakács, J.K. Paterson, J.A. Ludwig, C. Booth-Genthe, M.M. Gottesman, Nat. Rev. Drug Discov. 5 (2006) 219–234.
[4] A. Shafei, W. El-Bakly, A. Sobhy, O. Wagdy, A. Reda, O. Aboelenin, A. Marzouk, K. El Habak, R. Mostafa, M.A. Ali, M. Ellithy, Biomed. Pharmacother. 95 (2017) 1209–1218.
[5] G.S. Hassan, H.H. Georgey, E.Z. Mohammed, R.F. George, W.R. Mahmoud, F. A. Omar, Eur. J. Med. Chem. 218 (2021) 113389.
[6] M. Kim, M. Baek, D.J. Kim, Curr. Pharm. Des. 23 (2017) 4226–4246.
[7] R. Bansal, A. Malhotra, Eur. J. Med. Chem. 211 (2021) 113016.
[8] I.M.M. Othman, Z.M. Alamshany, N.Y. Tashkandi, M.A.M. GadElkareem, M. M. Anwar, E.S. Nossier, Bioorg. Chem. 114 (2021) 105078.
[9] M.F. Ahmed, E.Y. Santali, E.M.M. El-Deen, I.A. Naguib, R. El-Haggar, Bioorg. Chem. 106 (2021) 104473.
[10] J. Capdevila, E. Elez, T. Macarulla, F.J. Ramos, M. Ruiz-Echarri, J. Tabernero, Cancer treat, Rev 35 (4) (2009) 354–363.
[11] E.A. Abd El-Meguid, G.O. Moustafa, H.M. Awad, E.R. Zaki, E.S. Nossier, J. Mol. Struct. 1240 (2021) 130595.
[12] P. Seshacharyulu, M.P. Ponnusamy, D. Haridas, M. Jain, A.K. Ganti, S.K. Batra, Expert Opin. Ther. Targets 16 (2012) 15–31.
[13] R. Kannaiyan, D. Mahadevan, Expert Rev. Anticancer Ther. 18 (2018) 1249–1270.
[14] A. Petrelli, S. Giordano, Curr. Med. Chem. 15 (2008) 422–432.
[15] M.I. Ali, M.M. Naseer, RSC Adv. 13 (2023) 30462–30484.
[16] A. Sharma, R. Sheyi, B.G. de la Torre, A. El-Faham, F. Albericio, Molecules 26 (2021) 864.
[17] A. El-Faham, M. Farooq, Z. Almarhoon, R.M.A. Abd Alhameed, M. Wadaan, B.G. de la Torre, F. Albericio, Bioorg. Chem. 94 (2020) 103397.
[18] D. Kong, T. Yamori, Cancer Sci. 98 (2007) 1638–1642.
[19] J.R. Wong, L.M. Morton, M.A. Tucker, D.H. Abramson, J.M. Seddon, J.N. Sampson, R.A. Kleinerman, Am. J. Clin. Oncol. 32 (2014) 3284–3289.
[20] C.M. Dehnhardt, A.M. Venkatesan, Z. Chen, E. Delos-Santos, S. Ayral-Kaloustian, N. Brooijmans, K. Yu, I. Hollander, L. Feldberg, J. Lucas, R. Mallon, Bioorg. Med. Chem. Lett. 21 (2011) 4773–4778.
[21] E.S. Kim, Drugs 77 (2017) 1705–1711.
[22] N. Keldsen, H. Havsteen, I. Vergote, K. Bertelsen, A. Jakobsen, Gynecol. Oncol. 88 (2003) 118–122.
[23] Q. Dai, Q. Sun, X. Ouyang, J. Liu, L. Jin, A. Liu, B. He, T. Fan, Y. Jiang, Molecules 28 (2023) 4278.
[24] M. Zheng, C. Xu, J. Ma, Y. Sun, F. Du, H. Liu, L. Lin, C. Li, J. Ding, K. Chen, H. Jiang, Bioorg. Med. Chem. 15 (2007) 1815–1827.
[25] A.B. Patel, K.H. Chikhalia, P. Kumari, J. Saudi Chem. Soc. 18 (2014) 646–656.
[26] Q. Huang, Q. Fu, Y. Liu, J. Bai, Q. Wang, H. Liao, P. Gong, Chem. Res. Chinese Univ. 30 (2014) 257–265.
[27] J.K. Srivastava, G.G. Pillai, H.R. Bhat, A. Verma, U.P. Singh, Sci. Rep. 7 (2017) 5851.
[28] H.M.E. Marwa, I. Serag, M.G. Rania, M.A.M. Massoud, Design, Der Pharma Chem. 11 (2019) 7–14.
[29] Y.T. Abibigloo, S. Dastmalchi, N.R. Asl, J. Sh Mojarrad, Bioorg. Chem. 147 (2024) 107355.
[30] A. Sharma, H. Ghabbour, S.T. Khan, B.G. de la Torre, F. Albericio, A. El-Faham, J. Mol. Struct. 1145 (2017) 244–253.
[31] H.H. Khalil, M.M. El-Sheshtawy, Sh N. Khattab, M.M. Abu-Serie, M.G. Shehat, M. Teleb, N.S. Haiba, Bioorg. Chem. 144 (2024) 107155.
[32] M.H. El-Wakil, ShN. Khattab, A.F. El-Yazbi, N. El-Nikhely, A. Soffar, H.H. Khalil, Bioorg. Chem. 105 (2020) 104393.
[33] C. Li, F. Zhang, RSC Adv. 6 (2016) 75359–75364.
[34] G. Loloiu, T. Loloiu, O. Maior, Rev. Chim. (Buchar.) 49 (1998) 861–864.
[35] J.F.M. Da Silva, S.J. Garden, A.C. Pinto, J. Braz. Chem. Soc. 12 (2001) 273–324.
[36] N. Lashgari, G.M. Ziarani, ARKIVOC (Gainesville, FL, U. S.) (2012) 107–132, 2012.
[37] K.L. Vine, L. Matesic, J.M. Locke, M. Ranson, D. Skropeta, Med. Chem. 9 (2009) 397–414.
[38] L. Matesic, J.M. Locke, J.B. Bremner, S.G. Pyne, D. Skropeta, M. Ranson, K.L. Vine, Bioorg. Med. Chem. 16 (2008) 3118–3124.
[39] K. Han, Y. Zhou, F. Liu, Q. Guo, P. Wang, Y. Yang, B. Song, W. Liu, Q. Yao, Y. Teng, P. Yu, Bioorg. Med. Chem. Lett. 24 (2014) 591–594.
[40] H.S. Ibrahim, S.M. Abou-seri, N.S.M. Ismail, M.M. Elaasser, M.H. Aly, H.A. Abdel-Aziz, Eur. J. Med. Chem. 108 (2016) 415–422.
[41] H. Pervez, M. Ahmad, S. Zaib, M. Yaqub, M.M. Naseer, J. Iqbal, MedChemComm 7 (2016) 914–923.
[42] A.F.M.M. Rahman, S.-E. Park, A.A. Kadi, Y. Kwon, J. Med. Chem. 57 (2014) 9139–9151.
[43] Ganguly, S.; Debnath, B. 4 (2014) 558–568.
[44] M.F. Ahmed, R. El-Haggar, A.H. Almalki, O. Abdullah, M.A. El Hassab, N. Masurier, S.F. Hammad, Arch. Pharm. 356 (4) (2023) e2300244.
[45] H.E. Dweedar, H. Mahrous, H.S. Ibrahim, H.A. Abdel-Aziz, Eur. J. Med. Chem. 78 (2014) 275–280.
[46] W.M. Eldehna, A. Altoukhy, H. Mahrous, H.A. Abdel-Aziz, Eur. J. Med. Chem. 90 (2015) 684–694.
[47] H.S. Al-Salem, H.A. Abuelizz, I.S. Issa, A.Z. Mahmoud, A. AlHoshani, M. Arifuzzaman, A.F.M.M. Rahman, Appl. Sci. 10 (2020) 3669.
[48] L.T. Maillard, S. Bertout, O. Quinonéro, G. Akalin, G. Turan-Zitouni, P. Fulcrand, F. Demirci, J. Martinez, N. Masurier, Bioorg. Med. Chem. Lett. 23 (2013) 1803–1807.
[49] S. Murugappan, S. Dastari, K. Jungare, N.M. Barve, N. Shankaraiah, J. Mol. Struct. 1307 (2024) 138012.
[50] P. Nath, A. Mukherjee, S. Mukherjee, S. Banerjee, S. Das, S. Banerjee, Mini Rev. Med. Chem. 21 (2021) 1096–1112.
[51] G. Xu, M.C. Abad, P.J. Connolly, M.P. Neeper, G.T. Struble, B.A. Springer, S. L. Emanuel, N. Pandey, R.H. Gruninger, M. Adams, S. Moreno-Mazza, A.R. Fuentes-Pesquera, S.A. Middleton, Bioorg. Med. Chem. Lett. 18 (2008) 4615–4620.
[52] P. Kumar, B. Narasimhan, Mini-rev, Med. Chem. 13 (2013) 971–984.
[53] K. Hellmuth, S. Grosskopf, C.T. Lum, M. Würtele, N. Röder, J.P. von Kries, M. Rosario, J. Rademann, W. Birchmeier, Proc. Natl. Acad. Sci. USA 105 (2008) 7275–7280.
[54] J. Hofmann, J. Easmon, G. Puerstinger, G. Heinisch, M. Jenny, A.A. Shtil, M. Hermann, D.F. Condorelli, S. Sciré, G. Musumarra, Invest. New Drugs 27 (2009) 189–196.
[55] E.M. Fathalla, M.A.M. Abu-Youssef, M.M. Sharaf, A. El-Faham, A. Barakat, A.M. A. Badr, S.M. Soliman, A.M.Z. Slawin, J. Derek Woollins, Inorg. Chim. Acta. 555 (2023) 121586.
[56] H.H. Al-Rasheed, S.A. Al-khamis, A. Barakat, A. El-Faham, M. Haukka, S. M. Soliman, Crystals 13 (2) (2023) 305.
[57] I. Shawish, M.S. Nafie, A. Barakat, A. Aldalbahi, H.H. Al-Rasheed, M. Ali, W. Alshaer, M. Al Zoubi, S. Al Ayoubi, B.G. de la Torre, F. Albericio, A. El-Faham, Fornt, Chem 10 (2022) 1078163.
[58] I. Shawish, A. Barakat, A. Aldalbahi, W. Alshaer, F. Daoud, D.A. Alqudah, M. Al Zoubi, M.M. Hatmal, M.S. Nafie, M. Haukka, A. Sharma, B.G. de la Torre, F. Albericio, A. El-Faham, Pharmaceutics 14 (2022) 1558.
[59] I. Shawish, A. Barakat, A. Aldalbahi, A.M. Malebari, M.S. Nafie, A.A. Bekhit, A. Albohy, A. Khan, Z. Ul-Haq, M. Haukka, B.G. de la Torre, F. Albericio, ACS Omega 7 (2022) 24858–24870.
[60] K.S. Putt, G.W. Chen, J.M. Pearson, J.S. Sandhorst, M.S. Hoagland, J.-T. Kwon, S.-K. Hwang, H. Jin, M.I. Churchwell, M.-H. Cho, D.R. Doerge, W.G. Helferich, P. J. Hergenrother, Nat. Chem. Biol. 2 (2) (2006) 543–550.
[61] S.M. Maira, S. Pecchi, A. Huang, M. Burger, M. Knapp, D. Sterker, C. Schnell, D. Guthy, T. Nagel, M. Wiesmann, S. Brachmann, C. Fritsch, M. Dorsch, P. Chène, K. Shoemaker, A.D. Pover, D. Menezes, G. Martiny-Baron, D. Fabbro, C.J. Wilson, R. Schlegel, F. Hofmann, C. García-Echeverría, W.R. Sellers, C.F. Voliva, Mol. Cancer Ther. 11 (2012) 317–326.
[62] A.M. Venkatesan, Z. Chen, O.D. Santos, C. Dehnhardt, E.D. Santos, S.A. Kaloustian, R. Mallon, I. Hollander, L. Feldberg, J. Lucas, K. Yu, I. Chaudhary, T.S. Mansour, Bioorg. Med. Chem. Lett. 20 (21) (2010) 5869–5874.
[63] W.F. Zhu, Y.J. Liu, Y.F. Zhao, H.Y. Wang, L. Tan, W.J. Fan, P. Gong, Arch. Pharm. Chem. Life Sci. 345 (2012) 812–820.
[64] M. Zheng, C. Xu, J. Ma, Y. Sun, F. Du, H. Liu, L. Lin, C. Li, J. Ding, K. Chen, H. Jiang, Bioorg. Med. Chem. 15 (2007) 1815–1827.
[65] T. Żołek, A. Trzeciak, D. Maciejewska, J. Biomol. Struct. Dyn. 40 (2022) 622–634.
[66] H. Yang, Z. Zhang, Q. Liu, J. Yu, C. Liu, W. Lu, Molecules 28 (2023) 7607.
[67] C.H. Yun, T.J. Boggon, Y. Li, M.S. Woo, H. Greulich, M. Meyerson, M. Eck, J. Cancer Cell 11 (2007) 217–227.
[68] J. Sun, X.Y. Wang, P.C. Lv, H. Zhu, L. Sci. Rep. 5 (2015) 13934.
[69] J. Li, C. Ma, Y. Huang, J. Luo, C. Huang, Oncogene 22 (2003) 211–219.
[70] M.L. Martin-Fernandez, D.T. Clarke, S.K. Roberts, L.C. Zanetti-Domingues, F. L. Gervasio, Cells 8 (2019) 316.
[71] T. Amelia, R.E. Kartasasmita, T. Ohwada, D.H. Tjahjono, Molecules 27 (2022) 819.
[72] N.R. Patel, D.V. Patel, P.R. Murumkar, M.R. Yadav, Eur. J. Med. Chem. 121 (2016) 671–698.
[73] R.C.J. Becker, et al., Thrombolysis 52 (2021) 397–402.

[74] R. Lapatto, U. Krengel, H.A. Schreuder, A. Arkema, B. De Boer, K.H. Kalk, W.G. J. Hol, J.W.M. Mulders, R. Dijkema, EMBO J. 16 (1997) 5151–5161.
[75] W. Ma, C. Tang, L. Lai, Biophys. J. 89 (2005) 1183–1193.
[76] M.H. Abdel Aziz, P.S. Sidhu, A. Liang, J.Y. Kim, P.D. Mosier, Q. Zhou, D.H. Farrell, U.R. Desai, J. Med. Chem. 55 (15) (2012) 6888–6897.
[77] P.S. Sidhu, M.H. Abdel Aziz, A. Sarkar, A.Y. Mehta, Q. Zhou, U.R. Desai, J. Med. Chem. 56 (2013) 5059–5070.
[78] J.A.J. Huntington, et al., Haemostasis 3 (2005) 1861–1872.
[79] M.T. Nieman, F. Burke, M. Warnock, Y. Zhou, J. Sweigart, A. Chen, D. Ricketts, B. R. Lucchesi, Z. Chen, E. Di Cera, J. Hilfinger, J.S. Kim, H.I. Mosberg, A.H. J. Schmaier, et al., Haemostasis 6 (5) (2008) 837–845.
[80] A.E. Abd El-Wahab, D.A. Ghareeb, E.E. Sarhan, M.M. Abu-Serie, M.A. El Demellawy, Altern. Med. 13 (2013) 1–12.
[81] S. Umayaparvathi, S. Meenakshi, A. Muthuvel, T. Balasubramanian, Afr. J. Biotechnol. 12 (2013) 5897–5908.